Crossing the Color Line: Race, Place, and Deaths of Despair

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Abstract

Did the Great Migration and its consequences matter for later mortality? This paper shows that the mortality effects due to the rise in Chinese import competition in the early 21st century were mediated by the racial history of the most affected communities. In particular, the Great Migration of Blacks to Northern industrial cities between the 1940s and 1970s rendered those destinations with the highest migration flows more segregated and more vulnerable to economic shocks, resulting in higher levels of "deaths of despair" among both White and Black men. My findings show that the spatial vulnerabilities were not limited to Black communities, and they also challenge the conventional emphasis on "deaths of despair" among White men.

Keywords: Great Migration, China shock, deaths of despair, import competition, race, segregation.

JEL Codes: I14, I12, J15, F16, N32, R23.

1 Introduction

Deaths from drug overdoses, suicide, and alcoholic liver disease rose sharply among working-age American men in the 2000s. The leading account of these "deaths of despair" ties them to the erosion of stable jobs, predominantly for White men without a college degree (Case and Deaton, 2015; 2017). The timing of the rise in import competition fits this account: Chinese imports after 2000 reduced manufacturing employment in exposed labor markets (Autor et al., 2013; 2021; Pierce and Schott, 2016), and they have also been shown to raise deaths of despair (Autor et al., 2019; Pierce and Schott, 2020), as have other manufacturing shocks (Finkelstein et al., 2026). Racial decompositions show that Chinese import competition did particularly severe and long-lasting damage to White workers (Kahn et al., 2026). The resulting picture is that of a predominantly White working-class crisis.

This, however, leaves out the racial history of where these workers lived. Many of the local labor markets that Chinese competition hit were shaped by the Great Migration: between 1940 and 1970, about four million Black Americans left the South for cities in the North and the West. These destinations saw White flight, exclusionary zoning, residential segregation, incarceration, and a reduction in Black social mobility (Boustan, 2010; 2017; Derenoncourt, 2022; Ananat, 2011), as well as concentrated poverty and conflicts that shaped public goods provision (Massey, 1990; Trounstine, 2016). Thus, the racial history of places is tied to uneven consequences and has usually adversely affected Black Americans. Darity (2022) would also suggest that racial conflict could be beneficial for White workers.

The manufacturing employment decline could have gone either way for White workers in the more vulnerable places. If racial segregation and the institutions that grew up around it insulated White residents, high-migration destinations should have protected White workers from a shock that hit both Black and White workers. If instead this history made these places vulnerable for everyone, high-migration destinations should have been more vulnerable for Black and White residents. In this paper, I will show that this is the case.

I study 130 commuting zones outside the South and ask how the mortality effect of the 2000–2012 rise in Chinese import competition varied with the destination's historical exposure to the Great Migration. I look at both Black and White men aged 20–39. The

variable I am most interested in is the migration gradient: how much larger the effect of the same import shock is in a destination at the 75th percentile of predicted Great Migration relative to one at the 25th. I use the measure of predicted Great Migration from Derenoncourt (2022), which is built using southern push factors and pre-existing settlement networks, isolating a plausibly exogenous Black population increase. I instrument import exposure and its interaction with predicted migration with Chinese imports into other high-income countries.

My mortality data come from restricted-use death records for 2000–2013 that identify race as well as county of residence (National Center for Health Statistics, 2000). I particularly focus on two "deaths of despair" outcomes: accidental drug poisoning and a broader deaths of despair outcome that also includes suicide, alcoholic liver disease, and drug poisoning of undetermined intent.

My first finding is that the mortality effect of the same shock rose with predicted Great Migration exposure for both Whites and Blacks. Moving from the 25th to the 75th percentile of predicted migration increases the effect of a one-standard-deviation China shock by 105 deaths per 100,000 White men (SE = 39) and 82 per 100,000 Black men (SE = 41) over 2000–2013. This is about 7.5 and 5.8 deaths per 100,000 per year. These two gradients are not statistically distinguishable ($p = 0.41$), and the effects are largest among men with a high school education or less. I then show that this is not because of a larger shock: high-migration destinations were not systematically more exposed to import competition, and the shock's effect on manufacturing employment did not vary with predicted Great Migration. So, it is not a mechanically higher import shock in these places that was resulting in the higher mortality.

My second finding is that the mechanisms vary by race. For this, I use data from the Cooperative Election Study and various measures of race and class segregation using the Longitudinal Tract Database and the NHGIS. I find that among Black men, the gradient in accidental drug poisoning runs through concentrated poverty: the Great Migration's effect on the concentration of poverty in segregated neighborhoods accounts for about 35 to 45 percent of it. Poor Black residents of these destinations were isolated because racial segregation concentrated poverty in Black neighborhoods. Among White men, the gradient appears

across the whole commuting zone. White residents' opposition to redistributive spending accounts for about a sixth of the White gradient, although only marginally significantly, and is associated with smaller increases in local health spending after the China shock. The gradient is also associated, more weakly, with the residential separation of college-educated and non-college residents. Both Whites and Blacks were more vulnerable in these destinations, though through different channels.

This paper contributes to the existing Great Migration and segregation literature by demonstrating that the racial history of places matters for everyone, including Whites. It adds to that literature because most of the studies have found adverse effects for Black people (Derenoncourt, 2022; Ananat, 2011; Cox et al., 2026), with no clear consensus on the effects for Whites. For instance, Ananat (2011) finds that segregation increased metropolitan Black poverty and the Black–White income gap but reduced White poverty and inequality among White residents. My paper thus aligns closely with Chyn et al. (2026), who show that racial segregation reduced upward mobility for Black children and for low-income White children.

Also, these findings change how we think about the human cost of deindustrialization. The deaths-of-despair literature and the trade evidence that parallels it describe a crisis concentrated among White men without a college degree (Case and Deaton, 2017; Pierce and Schott, 2020; Kahn et al., 2026). The evidence here calls for rethinking that narrative: in the commuting zones restructured by the Great Migration and its aftermath, Black men's overdose deaths rose alongside White men's after the same shock. The comparison also speaks to stratification economics (Darity, 2022): in these destinations, residential segregation did not shield White residents from the rise in deaths of despair. I also add to the work on trade and racial gaps in health (Chatterji et al., 2026) by showing that the racial pattern depends on the history of places.

Finally, my paper contributes to the rethinking of the China shock literature, where recent discussions argue that while manufacturing job losses happened, the other aggregates such as wages and various other suggested mechanisms do not hold (Borusyak et al., 2022). Only looking at some of these economic aggregates understates the costs of trade. Work on NAFTA, the Great Recession, and import competition links local economic contractions to mortality and health (Finkelstein et al., 2026; 2025; Adda and Fawaz, 2020). And I show

that places with similar employment losses experienced very different mortality. Employment losses matter, but they are not enough to understand how places were harmed differently, and policy needs to reach the places whose history makes the same job losses more lethal.

2 Data

The main analysis combines four sources. The data assembled by Autor et al. (2021) provide the 2000–2012 change in Chinese import competition, the instrument, baseline characteristics, and population weights. Derenoncourt (2022) provides actual and predicted Great Migration exposure for 130 non-Southern commuting zones, together with the share of 1935–1940 Black migrants from the South. Mortality counts come from the restricted-use files of the CDC's National Center for Health Statistics, which cover every death certificate filed in the United States from 1990 to 2013 and record age, sex, race, education, and county of residence (National Center for Health Statistics, 2000). Annual SEER population estimates (Surveillance, Epidemiology, and End Results Program, 2026) provide the denominators for mortality rates. I focus on men ages 20–39, for comparability with Autor et al. (2019).

The Great Migration measure and its estimates are those of Derenoncourt (2022). Using the complete-count 1940 Census, Derenoncourt first maps recent Black migration networks from southern origin counties to northern destination cities. For each southern county j and destination city c, $\omega_{jc}^{1935-1940}$ is the share of Black migrants from j who lived in c in 1940 after living in j in 1935. These shares capture the tendency of later migrants to follow earlier migrants from the same communities. They are fixed before the 1940–1970 migration episode studied in the paper.

The origin-side shifts are predicted Black net migration from southern counties in the 1940s, 1950s, and 1960s. For each decade, a five-fold cross-validated LASSO selects among lagged southern push factors, including cotton and tobacco exposure, agricultural structure and tenancy, mining, and World War II defense-facility spending. The selected variables are then used in an ordinary least-squares prediction of county net migration. Predicted Black

inflow to destination c is

$$\widehat{\Delta B}_{c,1940-1970} = \sum_{j \in South} \omega_{jc}^{1935-1940} \widehat{m}_{j,1940-1970}, \tag{1}$$

where $\widehat{m}_{j,1940-1970}$ sums the three decadal predicted outflows from origin j. City predictions are aggregated to commuting zones, divided by the destination's 1940 total urban population, and converted to a percentile rank within the 130-zone sample. I call this predicted rank GM_c. Actual Great Migration exposure is the percentile rank of the realized 1940–1970 Black population growth, and the two ranks have a correlation of 0.527 in my sample.

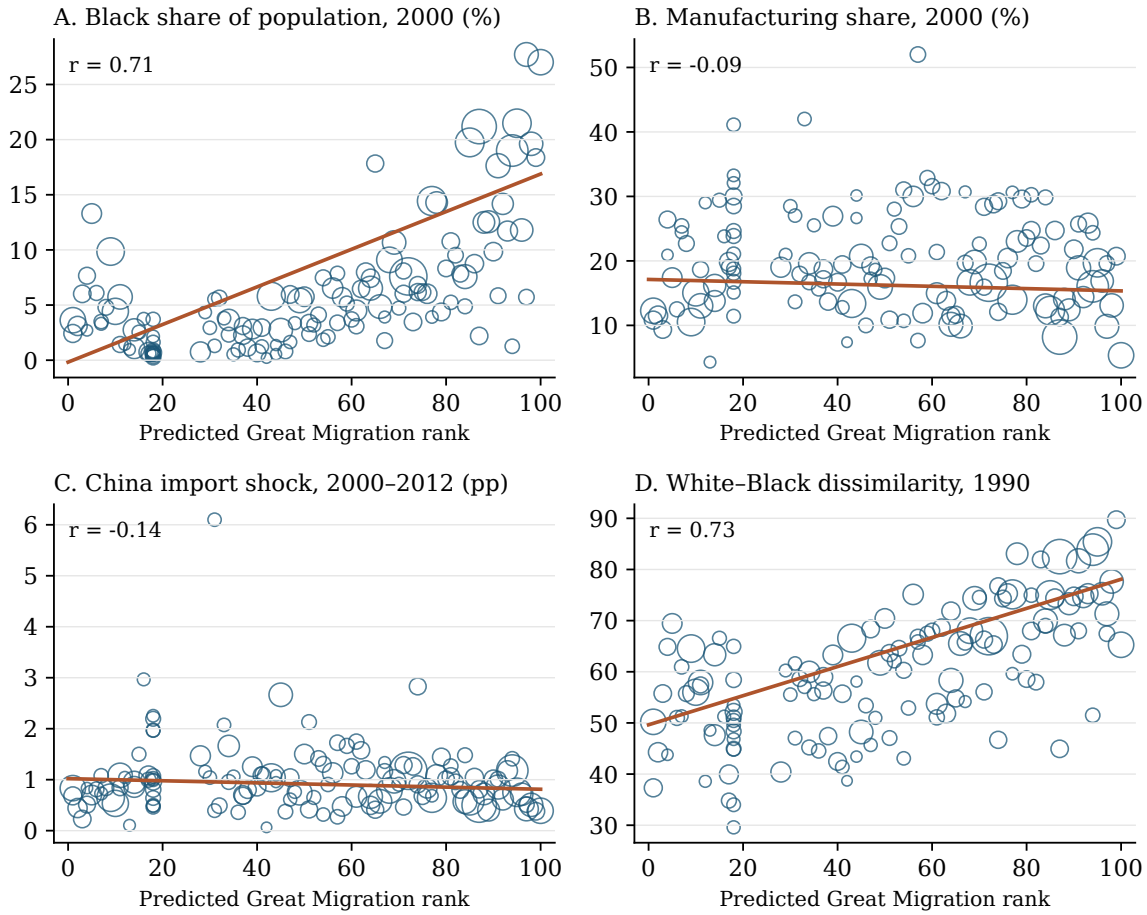
I use the predicted rather than the actual Great Migration, because the actual one has the endogeneity concerns that Derenoncourt (2022) points out. She finds that the actual 1940–1970 Black population increase is correlated with upward educational mobility, the manufacturing employment share, and the concentration of Black southern migrants in 1940. For my paper, the main concern is manufacturing. Places that were attractive to Black migrants may also have specialized in manufacturing. So, the effect of migration could be hard to distinguish from that of manufacturing job losses.

The other two correlations matter for the same reason. The destinations may have differed in labor demand and opportunities, which could also affect later mortality. I use the predicted measure to address these concerns. It combines southern-county push shocks, such as wartime defense spending and agricultural conditions, with migrant settlement networks from before 1940. Thus, they are plausibly unrelated to the pull factors in the destinations. The approach builds on the shift-share logic in Boustan (2010).

So, GM_c is the percentile rank of historically predicted Black population growth in commuting zone c. In the main specification, I use it as a predetermined moderator: I estimate how the effect of the China shock changes with GM_c, and I usually compare the effects at the 25th and the 75th percentiles of GM_c.

In Figure 1, I show how the measure relates to some characteristics of the destinations. Destinations with higher predicted migration had larger Black populations in 2000 (a population-weighted correlation of 0.71), and they were more segregated by race in 1990 (0.73). However, they did not have larger manufacturing sectors in 2000 (-0.09), and they were not more exposed to the China shock (-0.14).

Figure 1: Predicted Great Migration and the characteristics of destinations



Notes: Each circle is one of the 130 commuting zones in the mortality sample, plotted against its predicted Great Migration rank; circle areas increase with 2000 population. Lines are population-weighted linear fits, and r is the population-weighted correlation. No controls are applied. The 13 commuting zones at rank 18 have no predicted Black inflow and share a tied rank. Panel C is the 2000–2012 China import shock used throughout the paper (change in Chinese import penetration, percentage points, ten-year equivalent); Appendix Figure A1 shows the relationship after removing the regression controls. Panel D is the White–Black dissimilarity index across 1990 census tracts (0–100).
Sources: Autor et al. (2021); Derenoncourt (2022); Longitudinal Tract Database.

I study two mortality outcomes. The broader outcome follows the deaths-of-despair classification of Pierce and Schott (2020) and includes suicide, alcoholic liver disease, and accidental or undetermined-intent drug overdose. The narrower outcome, accidental drug poisoning, is a subset of it. Appendix Table A1 lists the ICD-10 codes for both.

Supplementary analyses use commuting-zone labor-market data, BRFSS health survey data, and neighborhood segregation measures computed from NHGIS and the Longitudinal Tract Database (Centers for Disease Control and Prevention, 2000; Schroeder et al., 2026;

Logan et al., 2014).

Mortality rates by commuting zone, year, sex, and race divide death counts by the corresponding annual SEER population and multiply by 100,000. The cumulative outcome sums these annual rates from 2000 to 2013, so the denominator follows the local population each year.

Table 1 describes the samples. The national frame has 722 commuting zones; predicted Great Migration exposure is available for the 130 commuting zones of the analysis sample. Compared with all commuting zones, the Great Migration sample has a similar manufacturing share and China-shock exposure, a lower Black population share because it excludes the South, and slightly higher college-educated and routine-occupation shares. Within the sample, top-quartile destinations have a Black population share 12 percentage points higher than bottom-quartile destinations, similar manufacturing shares, slightly lower exposure to the China shock, and higher all-cause mortality among young men in the 1990s. They were also more segregated and had more concentrated poverty in 1990.

Table 1 also reports both outcomes for White and Black men in the bottom and top quartiles of GM_c. In the raw averages, accidental drug poisoning among White men is much higher in top-quartile destinations (310 against 232 per 100,000), while the other three rates differ by 26 or less. These raw averages do not account for differences in exposure to the China shock, which the regressions below do.

3 Empirical specification

Let c index commuting zones and j industries. Following Autor et al. (2021), local exposure to the 2000–2012 China import shock is

$$X_c = 100 \sum_j s_{cj,2000} \, \Delta IP^{US}_{j,2000-2012}, \tag{2}$$

where $s_{cj,2000}$ is industry j's share of employment in commuting zone c at the start of the period, and $\Delta IP^{US}_{j,2000-2012}$ is the national increase in Chinese import penetration in that industry. Thus, a place is more exposed if it was initially specialized in the industries in which Chinese competition grew the most later on. I express the measure as a ten-year-

equivalent change, to keep it comparable to the earlier China-shock literature.

US imports could also have risen because of domestic demand. To isolate growth originating in Chinese supply, I instrument X_c with

$$Z_c = \sum_j s_{cj,1990}\, \Delta IP^{OHI}_{j,2000-2012}, \tag{3}$$

where $\Delta IP^{OHI}_{j,2000-2012}$ is the growth in Chinese exports to eight other high-income countries. The local employment shares are lagged by a decade. The idea behind this instrument is that increases in Chinese productivity and changes in market access would have also increased Chinese exports to other rich countries, which a demand change unique to the US would not do. Using lagged local industry shares further limits the possibility of anticipatory adjustments.

For each race-sex group r, the main mortality specification is

$$Y_{cr} = \alpha_r + \beta_{1r} X_c + \beta_{2r}(X_c \times GM_c) + \theta_r GM_c + \lambda_r(L_c \times GM_c) + W_c'\Gamma_r + \varepsilon_{cr}, \tag{4}$$

where Y_{cr} is cumulative 2000–2013 mortality per 100,000 men ages 20–39, GM_c is the percentile rank of predicted Great Migration exposure, and L_c is the initial manufacturing employment share. The coefficient β_{2r} measures whether the mortality effect of the China import shock is larger in commuting zones with more predicted Great Migration. Both X_c and $X_c \times GM_c$ are instrumented using Z_c and $Z_c \times GM_c$. Because GM_c moderates the shock, identifying β_{2r} requires more than the exogeneity of the China shock. The identifying assumption is that, conditional on the controls, predicted Great Migration exposure is not correlated with other destination characteristics that make mortality more sensitive to an economic shock. Using predicted migration addresses the endogeneity of where the migrants settled, but it does not address every characteristic that is correlated with that prediction. I test this assumption in two ways: in Section 5, I add employment-composition controls and randomize the migration assignment, and in Section 6, I interact the shock with other destination characteristics.

Following Autor et al. (2021), I include the 2000 manufacturing employment share, college-educated and foreign-born population shares, female employment share, routine-intensive employment share, offshorability measure, population shares ages 0–17, 40–64,

and 65 and older, and Census division fixed effects. I also control for predicted Great Migration rank, the sum of lagged industry exposure shares, and the interactions of predicted migration with the manufacturing employment share and the exposure-share sum. Following Derenoncourt (2022), I include the share of the 1940 urban population consisting of Black migrants from southern counties who arrived during 1935–1940. Finally, I control for cumulative all-cause mortality among men ages 20–39 over 1990–1999. I do not control for the contemporary racial composition, as it is partly an outcome of the Great Migration. The regressions are weighted by the commuting-zone population in 2000.

With this specification, I can look at three different hypotheses. First, a *White-predominant mortality account* would predict no response in Black male mortality in the high-GM_c destinations, and I test this by asking whether $\beta_{1,BM} + g_H\beta_{2,BM} = 0$, where g_H is the 75th percentile of predicted Great Migration exposure. The *racialized-place mortality hypothesis* predicts $\beta_2 > 0$. The conditions produced by the Great Migration would make a later economic shock more damaging for both White and Black residents. On the other hand, an account around dominant group advantages would suggest that Whites were comparatively insulated in these places. I would have to reject $\beta_{2,BM} = \beta_{2,WM}$, to demonstrate that the slopes are different. Thus, my main evidence for shared vulnerability is the gradient: a positive $\beta_{2,BM}$ that I cannot distinguish from $\beta_{2,WM}$. The test of $\beta_{1,BM} + g_H\beta_{2,BM} = 0$ is secondary, because the level effect at a single percentile is estimated less precisely.

Following Borusyak et al. (2022), I compute and report the standard errors at the industry level. For the interaction, the same industry shocks are additionally weighted by predicted Great Migration exposure. I estimate the Black and White equations jointly when I compare their coefficients. To make the estimates easier to interpret, I report the effects of a one-standard-deviation China import shock at the 25th, 50th, and 75th percentiles of predicted Great Migration exposure, along with the difference between the 75th and 25th percentiles.

4 Results

4.1 Mortality effects across Great Migration destinations

In Table 2, I report the main results for accidental drug poisoning (Panel A) and the broader deaths of despair outcome (Panel B). The China shock is scaled to one standard deviation, and the outcomes are cumulative deaths per 100,000 men ages 20–39 over 2000–2013. Columns 1, 3, and 5 report the average effect of the China shock, without the Great Migration interaction. Among all men, the average effect is 17.7 for accidental poisoning (SE = 18.2) and 31.2 (SE = 17.7) for the broader outcome. For the broader outcome, the White and Black estimates are similar in size (38.5 and 33.7); for accidental poisoning, the White estimate is larger (22.2 against 9.2). The Black estimates are less precise.

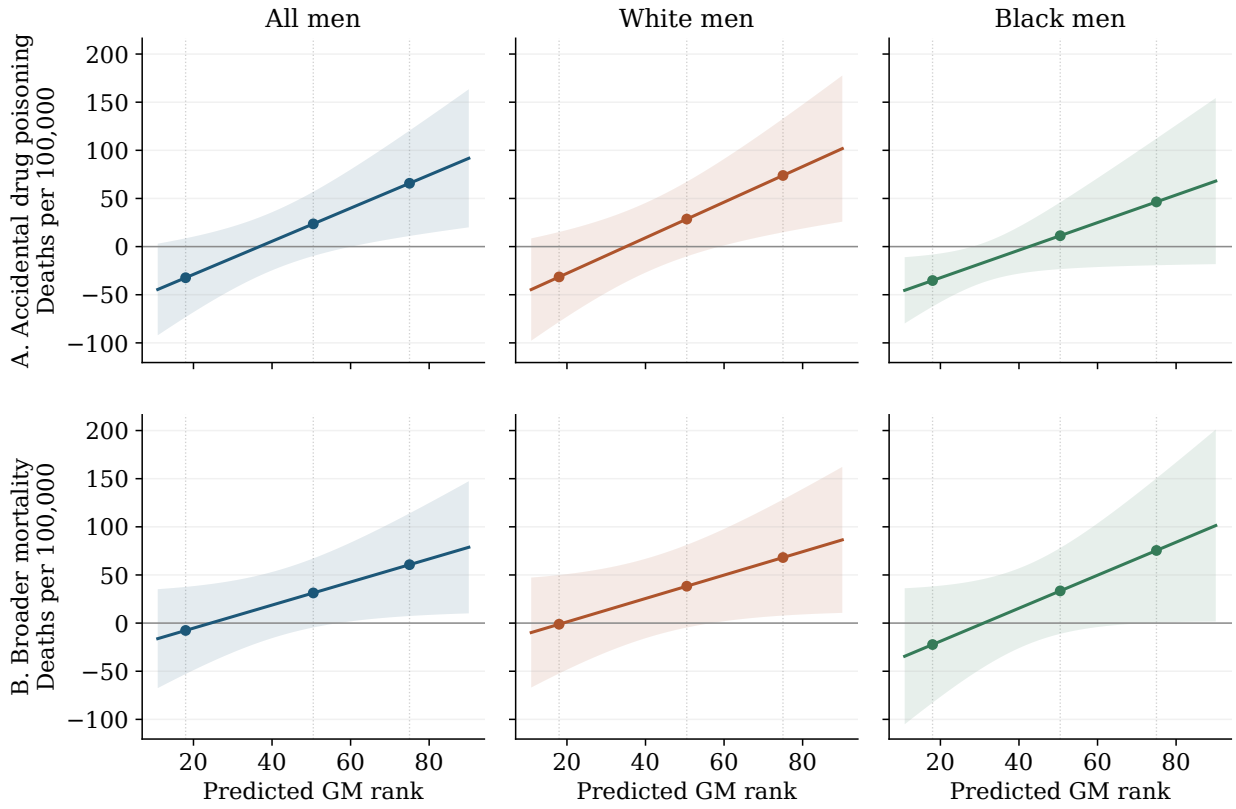
The even-numbered columns ask whether the response is larger in destinations with a larger predicted Black population increase during the Great Migration. Because GM_c is centered at its median, the uninteracted China-shock coefficient is the effect of a one-standard-deviation shock at the median predicted rank, and these estimates are close to the average effects. Moving from the 25th to the 75th percentile of GM_c raises the broader-outcome effect by 68.3 (SE = 35.9) among all men, 69.3 (SE = 37.9) among White men, and 97.7 (SE = 54.4) among Black men. For accidental poisoning, the gradients are more precisely estimated and, for all men and White men, larger: 98.1 (SE = 37.4) among all men, 105.4 (SE = 38.8) among White men, and 81.7 (SE = 41.0) among Black men. Appendix Table A2 splits the broader outcome by cause. Its gradient comes from accidental drug poisoning: the gradients for undetermined-intent drug poisoning, suicide, and alcoholic liver disease are small and not statistically significant, and for White men the first two are negative.

Figure 2 plots the fitted effect across the predicted-migration distribution, and Appendix Table A3 reports the estimates. At the 75th percentile, a one-standard-deviation China shock raises broader mortality by 60.7 (SE = 27.2) among all men, 68.1 (SE = 30.8) among White men, and 75.4 (SE = 38.2) among Black men, or 10.9, 11.1, and 18.7 percent of the respective cumulative means. For accidental poisoning, the corresponding effects are 65.8 (SE = 27.9), 73.9 (SE = 30.1), and 46.4 (SE = 33.6); the Black effect at high migration is positive but not statistically significant.

Table 3 tests whether the gradients differ by race, estimating Black and White equations jointly. The Black–White difference is 28.4 (SE = 39.6) for the broader outcome ($p = 0.472$) and -23.7 (SE = 28.6) for accidental poisoning ($p = 0.407$). The gradient is positive for both White and Black men, and the two cannot be distinguished statistically. The estimates are therefore inconsistent with a White-predominant mortality response, but they do not show that the gradient differs by race.

Autor et al. (2019) use a related but distinct poisoning outcome, in a stacked specification that includes the 1990s and focuses on male–female differences. With their poisoning measure in this paper's specification, the effect at the 75th percentile of predicted Great Migration is 49.3 deaths per 100,000, against a sample-average effect of 18 (Appendix Table A4).

Figure 2: Mortality effects of Chinese import competition across Great Migration destinations, 2000–2013



Notes: Curves show the fitted effects of a one-standard-deviation increase in Chinese import exposure from the interacted models in Table 2. Outcomes are cumulative deaths per 100,000 men ages 20–39 over 2000–2013, calculated by summing annual death rates. The top row shows accidental drug poisoning; the bottom row adds suicide, alcoholic liver disease, and drug poisoning of undetermined intent. Columns distinguish all men, White men, and Black men. The sample contains 130 commuting zones. Curves span the 10th–90th percentiles of predicted Great Migration rank; points and vertical dotted lines mark the 25th, 50th, and 75th percentiles. Shaded areas are pointwise 95 percent confidence intervals, calculated from the full coefficient covariance matrix using exposure-robust standard errors clustered by three-digit industry.

4.2 Import exposure and employment responses across Great Migration destinations

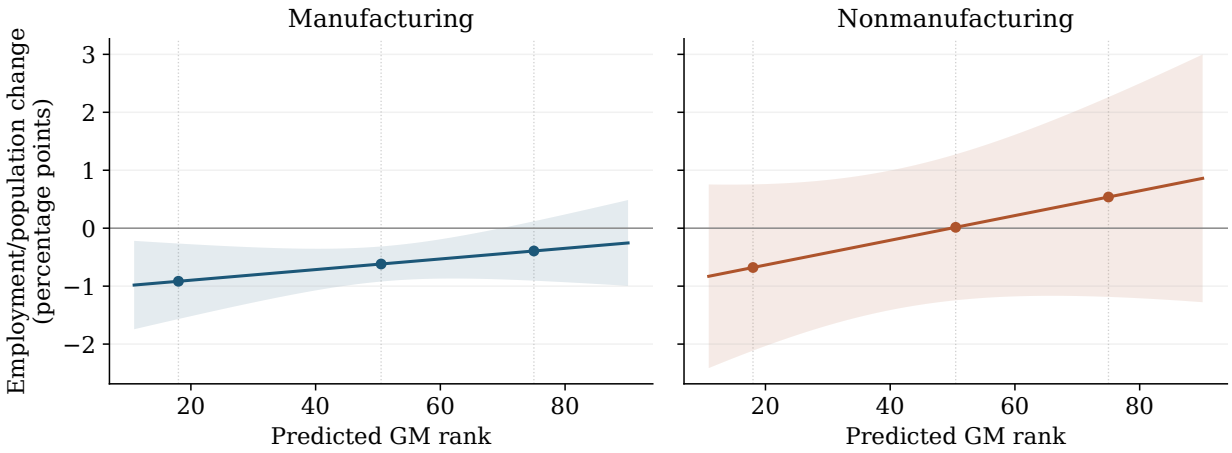
Did destinations with more predicted Great Migration also face more Chinese import competition, or lose more manufacturing jobs when it arrived?

Appendix Figure A1 plots observed China-shock exposure against predicted Great Migration rank in the analysis sample. The population-weighted raw correlation is −0.138; after residualizing both variables on the regression controls, it is 0.127 (Appendix Table A5). With equal weights across commuting zones, the raw and control-adjusted correlations are

−0.086 and 0.045. Predicted Great Migration and China-shock exposure are thus only weakly associated, before and after adjustment: high-migration destinations did not face a systematically larger import shock.

For the second question, I compare the estimated effects of import competition on employment in the Great Migration sample with the effects across all commuting zones (Appendix Figure A2). Both samples show adverse manufacturing employment effects, so the manufacturing contraction is relevant to my restricted sample. Then, in Table 4, I allow the employment response to vary with the predicted Great Migration rank. At the median rank, a one-standard-deviation China shock reduces manufacturing employment by 0.618 percentage points (SE = 0.155). Moving from the 25th to the 75th percentile of predicted Great Migration changes this effect by +0.523 percentage points, which is not statistically significant (SE = 0.505). Figure 3 shows the fitted employment effects across the predicted-migration distribution. Together, the exposure correlations and the employment estimates do not support an explanation in which the larger mortality response reflects greater import exposure or larger manufacturing job losses in high-migration destinations. Within the Great Migration sample, the shock lowered manufacturing employment among both White and Black residents, and the Black–White difference is not statistically significant (Appendix Table A6).

Figure 3: Employment responses across Great Migration destinations, 2001–2012



Notes: Curves show the fitted effects of a one-standard-deviation increase in Chinese import exposure on the 2001–2012 change in employment relative to the working-age population. Effects are in percentage points. The left panel shows manufacturing employment and the right panel nonmanufacturing employment. Estimates use the interacted models in Table 4 and the 130-zone Great Migration sample. Curves span the 10th–90th percentiles of predicted Great Migration rank; points and vertical dotted lines mark the quartiles. Shaded areas are pointwise 95 percent confidence intervals, calculated from the full coefficient covariance matrix using exposure-robust standard errors clustered by three-digit industry.

4.3 Education and the social gradient in mortality

The literature on deaths of despair (Case and Deaton, 2015; 2017) emphasizes the mortality and morbidity of White men without a college degree, and Finkelstein et al. (2026) find that the mortality effects of manufacturing shocks fall mainly on less-educated men. Educational attainment may itself have been shaped by the Great Migration and its social consequences (Boustan, 2017). It is therefore useful to ask whether the gradient holds within education groups: were high-migration destinations more dangerous for less-educated men than low-migration destinations were for less-educated men, or does the gradient simply reflect more less-educated men in high-migration destinations?

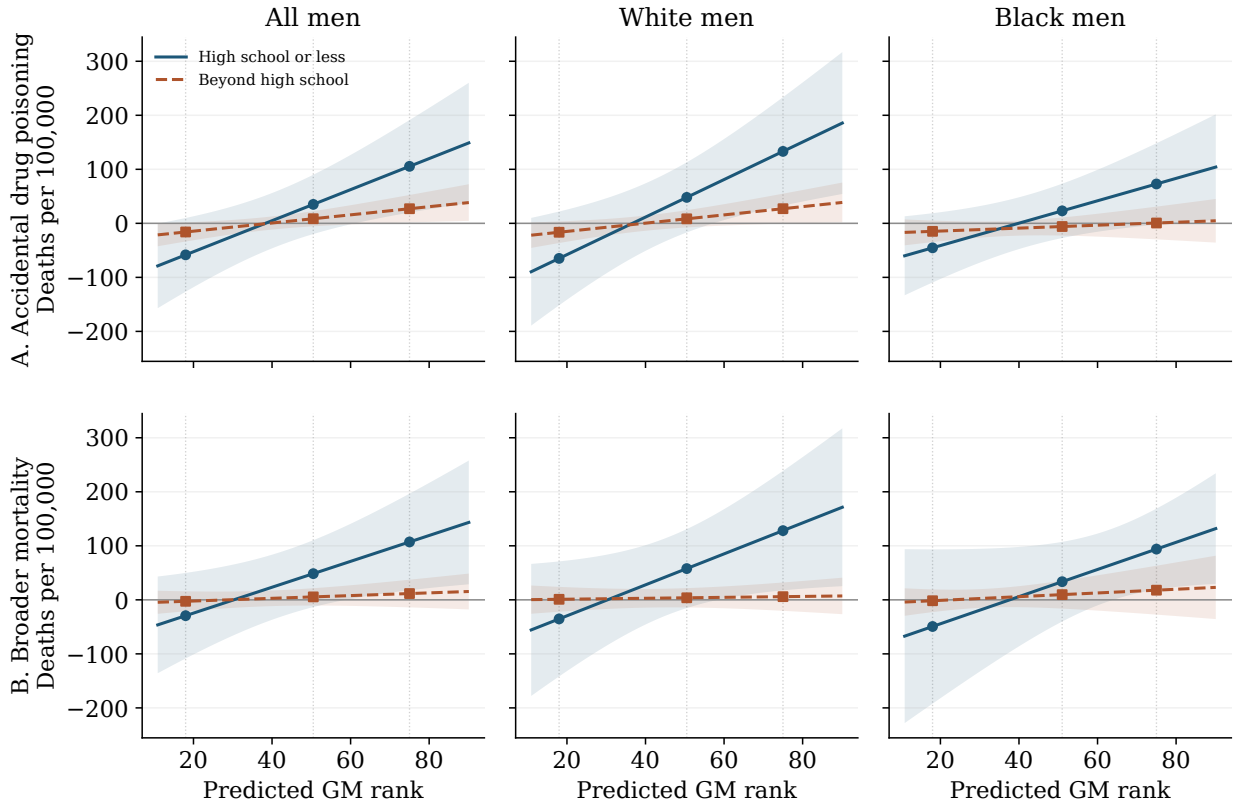
This exercise requires two assumptions. First, the main analysis uses annual population estimates by age, but population by education is available only from the decennial census, for broader age groups. I divide men ages 20–39 into those with a high school education or less and those with more, using 1990 NHGIS male education shares, and compute mortality for each group; this assumes that education composition does not change over the mortality window. Second, deaths with unknown education are allocated in proportion to the deaths

with known education within each commuting zone, race, and year. The pattern of results is not sensitive to this allocation.

Figure 4 and Appendix Tables A7 and A8 show that the increase in deaths of despair is concentrated among men with a high school education or less. At the 75th percentile of predicted Great Migration exposure, the effects on accidental drug poisoning in this group are larger than in the pooled sample: 105.5 deaths per 100,000 among all men (SE = 43.7), 133.2 among White men (SE = 51.2), and 72.9 among Black men (SE = 38.5). The effects beyond high school are smaller but positive: 27.1 among all men (SE = 13.0), significant at the 5 percent level, and 27.1 among White men (SE = 14.2), significant at the 10 percent level (Appendix Table A9, Panel B). The interaction coefficient is positive and significant for all, White, and Black men with a high school education or less, and the broader outcome shows the same pattern. Table 5 tests the education contrast directly. At the 75th percentile, the effect of the shock on accidental drug poisoning among men with high school or less exceeds the effect among men with more education by 78.5 deaths per 100,000 among all men, 106.0 among White men, and 72.2 among Black men; the difference is significant for each group and for both outcomes. As the education-specific results depend on the proportional allocation, I show in Appendix Tables A9 and A10 that excluding the unknown-education deaths keeps the education contrast, although some effect magnitudes and significance levels change.

Thus, the within-education results show that education composition alone is not enough to explain the mortality gradient that I find. The effects are concentrated among less-educated men, but less-educated men in high-migration destinations were more vulnerable to the same shock than less-educated men elsewhere.

Figure 4: Education differences in mortality responses across Great Migration destinations, 2000–2013



Notes: Curves show the fitted effects of a one-standard-deviation China shock on cumulative deaths per 100,000 men ages 20–39 over 2000–2013. Solid lines represent men with high school or less; dashed lines represent men with education beyond high school. The top row shows accidental drug poisoning; the bottom row adds suicide, alcoholic liver disease, and drug poisoning of undetermined intent. All-men and White-male estimates use 130 commuting zones; Black-male estimates use 129. Deaths with unknown education are allocated proportionally, and fixed 1990 education shares divide annual population denominators, as in Appendix Tables A7 and A8. Curves span the 10th–90th percentiles of predicted Great Migration rank in the main sample; markers show each equation's quartiles. Shaded areas are pointwise 95 percent confidence intervals, calculated from the full coefficient covariance matrix using exposure-robust standard errors clustered by three-digit industry.

5 Robustness and identification

This section shows that the results are robust to changes in population composition, to placebo migration assignments, and to controls for occupational, health, and education composition, and it checks the pattern with a different identification strategy based on permanent normal trade relations.

5.1 Population composition and outcome measurement

Mortality rates divide death counts by the annual SEER population. If the China shock caused larger net population losses in high-migration destinations, the shrinking denominators could create the appearance of a mortality gradient.

Appendix Table A11 reports the effect of the shock on the 2000–2012 change in log population at ages 18–39. The Great Migration gradient is positive but not statistically significant, and the same holds for all men, White men, and nonwhite men separately. The one significant gradient, for men ages 18–24 (4.0, SE = 1.8), has the opposite sign from the concern: faster population growth in high-migration destinations would lower measured mortality rates there and work against the mortality gradient. I do not observe gross flows, however, and cannot rule out that movers and stayers differed in underlying health.

5.2 Placebo migration assignments

Following the randomization-inference exercise in Derenoncourt (2022), I randomly reassign predicted out-migration across southern origin counties 1,000 times and combine each assignment with the historical migration links to compute predicted exposure for each destination. I then re-estimate the mortality specification for each of the 1,000 assignments, replacing predicted migration throughout the model.

For the broader outcome, none of the 1,000 assignments produces a structural interaction statistic as large as the observed one (randomization $p = 0.001$); for accidental drug poisoning, 6 do ($p = 0.007$). The corresponding reduced-form p-values are 0.009 and 0.039 (Appendix Table A12). The actual migration assignment produces far stronger mortality heterogeneity than random assignments do.

5.3 Employment composition: education, health, and occupations

Even with similar lagged manufacturing shares, the China shock may have been buffered in commuting zones with larger education or health-care sectors, and predicted Great Migration rank could partly capture such differences in employment composition. I therefore add industry and occupation controls to the mortality specification.

I add industry and occupation shares separately and together. Educational services and health care and social assistance are industries; occupation shares capture the type of work performed across industries. The shares come from Census 2000 SF3 counts aggregated from counties to commuting zones and are expressed as shares of employed civilian residents ages 16 and older, of both sexes. The 1990 census provides a sensitivity check.

Appendix Table A13 reports the effects at the 75th percentile of predicted migration and the 75th-minus-25th-percentile gradients. Education and health-sector shares do not attenuate the White estimates: with both shares, the effects at the 75th percentile are 73.8 for accidental drug poisoning and 70.0 for the broader outcome. Occupational composition matters more: 2000 occupation shares lower these effects to 43.4 and 41.3. The Black accidental-poisoning gradient is more sensitive to industry composition: education and health shares lower it from 81.7 to 63.2.

I do not find the changes to be monotonic across the control sets. When I control for both industry and occupation shares, the White effects at the 75th percentile are 52.6 and 64.7, although the broader mortality gradient increases compared to the baseline. Paired comparisons show that these changes are usually imprecisely estimated. One exception is White accidental poisoning, where the 1990 occupation controls reduce the effect at the 75th percentile to 54.6, and this change is significant at the 5 percent level. These controls also weaken the first stage (the Kleibergen–Paap F falls from 35.16 to 11.58).

So, while I do find some attenuation, it is possible that adding occupation shares on top of the industry controls reduces power in a sample of 130 commuting zones. Moreover, even though these differences could explain some of the effects, most of the effects at the 75th percentile and most of the interaction terms remain positive and statistically significant.

5.4 PNTR exposure and the timing of the mortality gradient

Permanent Normal Trade Relations (PNTR) with China, which was enacted in 2000 and came into effect after China's accession to the WTO in 2001, removed the possibility that China's access to the low U.S. Normal Trade Relations (NTR) tariff schedule could be revoked. In doing so, it gives me an alternative identification strategy. Pierce and Schott (2016) exploit cross-industry differences in the resulting policy uncertainty, which they mea-

sure as the gap between the statutory column 2 tariff and the NTR tariff. Industries with a larger NTR gap had a larger reduction in uncertainty after 2001, when the possibility of reverting to the higher tariffs was eliminated. Thus, in this design, the predetermined industry exposure is interacted with the post-2001 period, and, following Pierce and Schott (2016), I aggregate the industry-level shock to commuting zones by weighting the industry gaps with each commuting zone's pre-PNTR employment shares.

Let P_c denote local PNTR exposure and G_c predicted Great Migration rank, each centered at its unweighted mean and divided by its interquartile range in the 130-CZ frame. For race group r, the event-study specification is

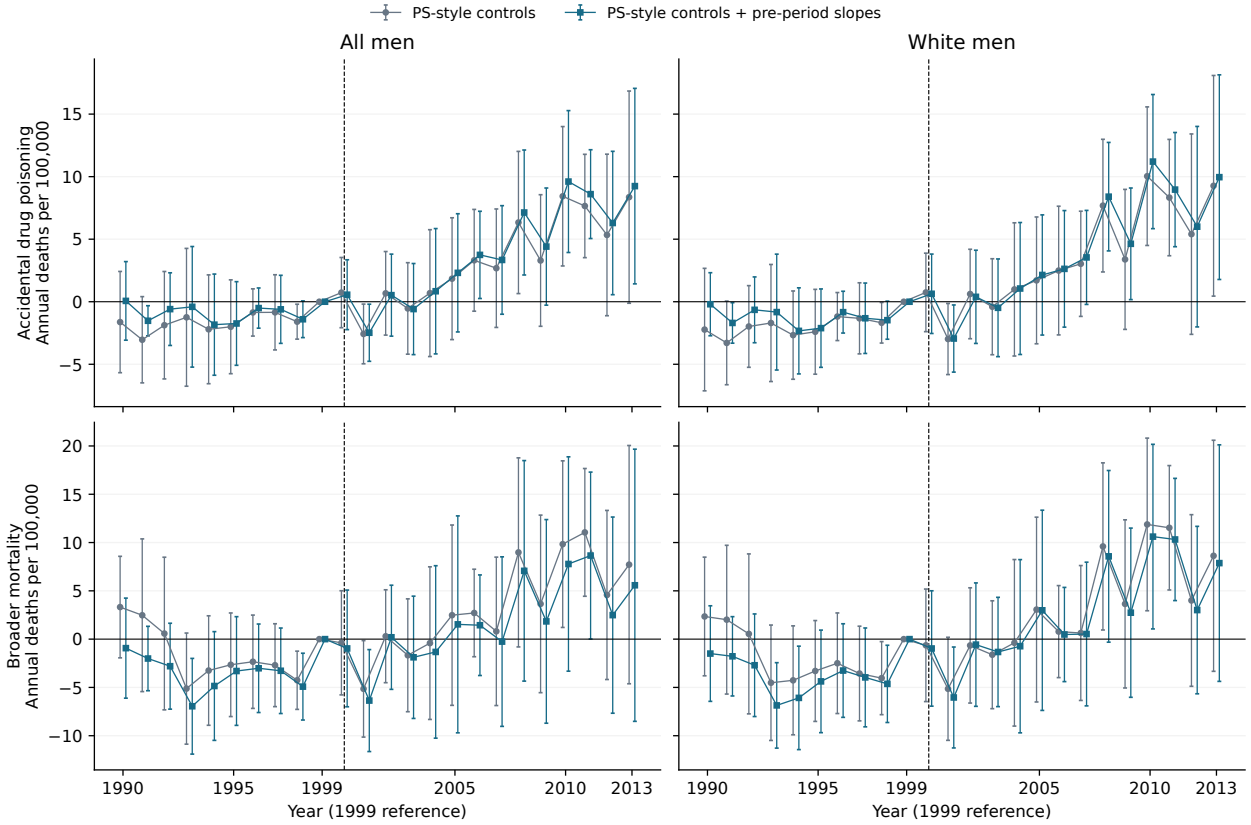
$$y_{crt} = \alpha_{cr} + \lambda_{rt} + \sum_{\tau \neq 1999} \mathbf{1}\{t = \tau\} \left[\beta_{r\tau} P_c + \gamma_{r\tau} G_c + \delta_{r\tau} P_c G_c + W_{ct}' \pi_{r\tau}\right] + Q_{ct}' \eta_r + \varepsilon_{crt}. \tag{5}$$

I estimate this specification separately for annual mortality rates from the broader and narrower outcomes, over 1990–2013 with 1999 as the reference year, and with commuting-zone and year fixed effects specific to each group. The coefficient $\delta_{r\tau}$ measures the change, relative to 1999, in the mortality gradient: the effect of a one-IQR increase in PNTR exposure across one IQR of predicted Great Migration. All lower-order terms are included.

W_{ct} contains time-invariant controls interacted with year indicators: exposure to changes in Chinese tariffs and subsidies, the veteran and foreign-born shares, and the initial manufacturing share. The policy controls Q_{ct} vary over time and enter directly. An alternative specification also controls for pre-period mortality trends.

I report the interaction coefficients $\delta_{r\tau} - \delta_{r,1999}$ for age-standardized annual rates among all, White, and Black men. Figure 5 plots the annual interaction coefficients for men ages 20–39. Averaged over 2008–2013 relative to 1999, the accidental-poisoning gradient is 6.57 deaths per 100,000 among all men (SE = 2.47) and 7.35 among White men (SE = 2.50). The Black estimates show strong pre-trends and, as in Pierce and Schott (2020), cannot be used; the broader outcome also shows substantial pre-trends. Appendix Table A14 reports other age groups. The PNTR evidence is nonetheless suggestive for accidental drug poisoning among young men, and it comes from a different source of variation in Chinese competition over the same period.

Figure 5: Trade liberalization and mortality across Great Migration destinations, 1990–2013



Notes: Points show changes in the annual mortality response to permanent normal trade relations exposure across one interquartile range of predicted Great Migration, per interquartile range of tariff-gap exposure, relative to 1999. The sample contains 130 commuting zones over 1990–2013. Outcomes are age-standardized deaths per 100,000 men ages 20–39, using common national population shares in 2000. Accidental drug poisoning is the narrow outcome; broader mortality also includes suicide, alcoholic liver disease, and drug poisoning of undetermined intent. Gray circles show the Pierce–Schott-control adaptation described in the text; blue squares add outcome- and group-specific pre-period mortality slopes interacted with year. Both models include commuting-zone and year effects, both lower-order exposure-by-year terms, and fixed 1990 group-population weights. Bars show pointwise 95 percent intervals clustered across 32 states; inference with slope controls conditions on the estimated slopes. The dashed line marks 2000, the year PNTR was granted. Points are offset for legibility. Appendix Table A14 reports pretrend tests and age extensions.

6 Mechanisms: the Great Migration, segregation, and concentrated poverty

So far, I have shown that Chinese import competition in the 2000s raised deaths of despair more in high-migration destinations. This section asks how the Great Migration changed these destinations. Derenoncourt (2022) shows that racial segregation and incarceration rose after the Great Migration, but finds no differences in public goods provision, possibly

because the outcomes are aggregated to the commuting zone. At the metropolitan level, racial residential segregation has been shown to reduce public goods provision (Cox et al., 2026). The links between the Great Migration and class segregation, and their consequences, are less clear: Boustan (2017) argues that zoning was used to exclude both poor Black and poor White residents, while Derenoncourt (2022) finds little evidence of income-based segregation.

Racial and class segregation suggest different mechanisms through which high-migration destinations could have become more vulnerable. Trounstine (2016) argues that racial groups in segregated cities are more divided over political priorities, so that diverse but divided cities find it harder to agree on basic government services and provide fewer public goods. Massey (1990) places racial segregation at the center of the inner-city crisis; in that account, concentrated poverty makes Black residents more vulnerable than White residents. Wilson (1991) emphasizes the social isolation of the "underclass," which deprives residents of resources as well as of social networks and role models.

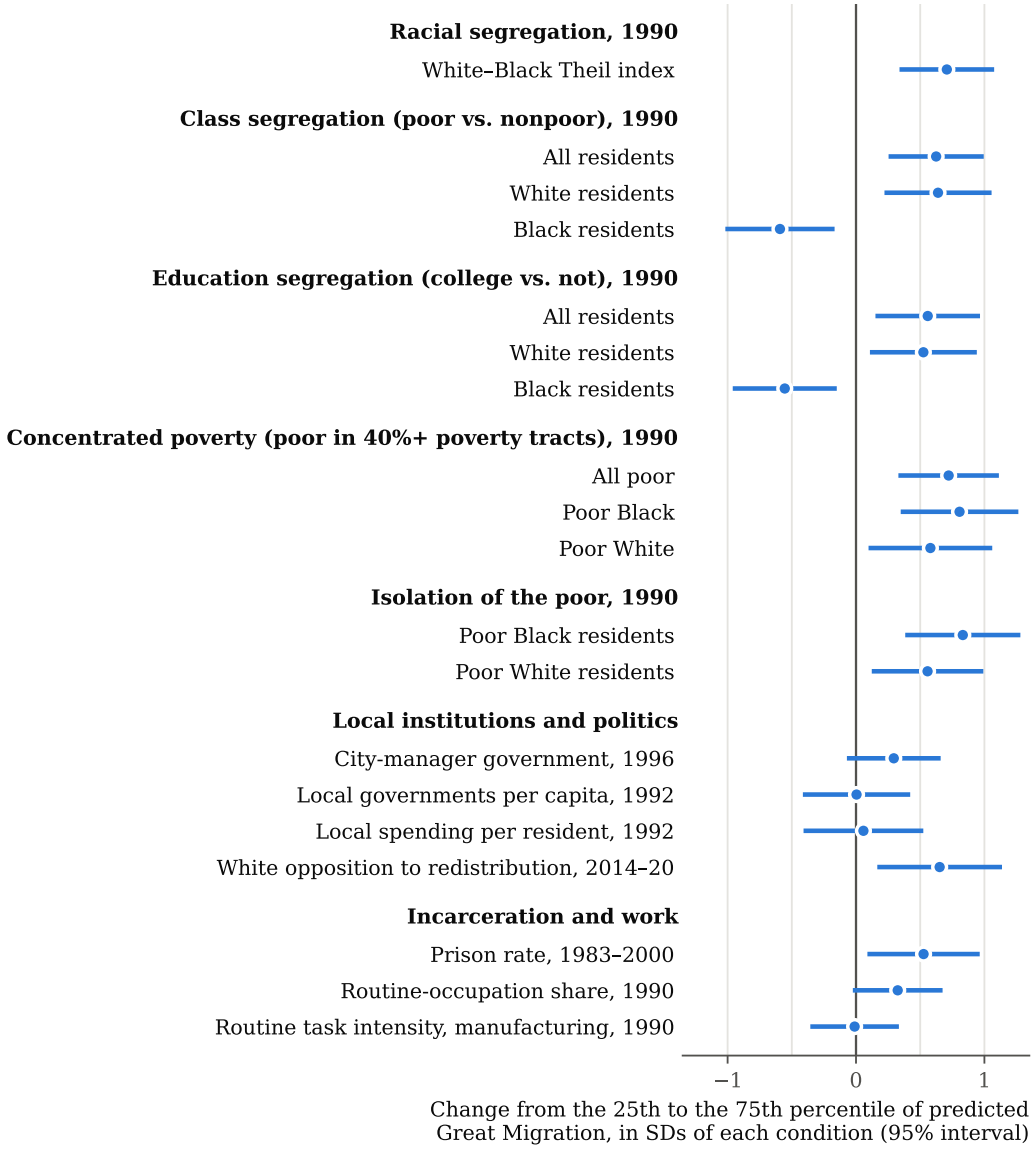
The section is organized around three hypotheses, one for each account. Each makes two claims: that the Great Migration changed a particular feature of its destinations, and that this feature made the China shock deadlier. Section 6.1 describes how I test the two claims, Sections 6.2–6.4 take the hypotheses in turn, Section 6.5 considers other features of the destinations, and Section 6.6 summarizes which channels operate, for whom, and how much of the Great Migration gradient they account for.

6.1 The design

Many features of local labor markets, such as poverty, segregation, and isolation between and within groups, changed after the Great Migration. For each feature S_c, measured before the China shock, I first regress S_c on predicted Great Migration rank with Derenoncourt (2022)'s controls: 1940 educational upward mobility, the 1940 manufacturing share of the labor force, the share of 1940 urban residents who were Black migrants from the South in 1935–1940, and Census region indicators. Commuting zones are weighted equally, and standard errors are heteroskedasticity robust. Figure 6 reports the implied difference between destinations at the 25th and 75th percentiles of predicted migration, in standard deviations of the feature.

High-migration destinations were more segregated by race. All residents, and White residents in particular, were more segregated by class and by education; poverty was more concentrated, and poor residents of both races were more isolated from nonpoor neighbors. Incarceration and White opposition to redistributive spending were also higher. Black residents, by contrast, were less segregated by class and education. I find no differences in local fiscal capacity or local government fragmentation at the commuting-zone level. Appendix Table A15 reports instrumental-variable estimates of the effect of actual migration on segregation and concentrated poverty, with predicted migration as the instrument; they point the same way.

Figure 6: How high- and low-migration destinations differed before the China shock



Notes: Each row reports the difference in a destination feature between commuting zones at the 75th and 25th percentiles of predicted Great Migration rank, in standard deviations of the feature, with a 95 percent confidence interval. Estimates come from regressions of each feature on predicted migration rank with Derenoncourt (2022)'s controls (1940 educational upward mobility, 1940 manufacturing labor-force share, 1935–1940 Black southern-migrant share, and Census region indicators), with heteroskedasticity-robust (HC3) standard errors; 130 commuting zones unless the feature is missing. Each label gives the year in which the feature is measured. Theil indices measure segregation between poor and nonpoor persons (class), adults with and without a four-year college degree (education), and White and Black persons (race). Concentrated poverty is the share of poor residents living in tracts with poverty rates of 40 percent or more; isolation is the poor share among the neighbors of the average poor resident. City-manager government is the population-weighted share of covered municipal residents under city-manager government (Municipal Yearbook data of Grumbach et al. 2026). Opposition to redistribution is the CCES index for White respondents, 2014–2020. *Sources:* LTDB and NHGIS tract data; Vera Institute incarceration data; Census of Governments; Adda and Fawaz (2020); Cooperative Congressional Election Study.

I then ask whether these features made the shock deadlier. For each feature S_c, I add to the mortality model the interaction of the China shock with S_c, instrumented in the same way, and I control for S_c and its interactions with the manufacturing share and the sum of industry shares:

$$Y_c = \beta_1 X_c + \beta_2 (X_c \times GM_c) + \beta_3 (X_c \times S_c) + W_c'\gamma + \varepsilon_c. \tag{6}$$

The coefficient β_3 measures whether the China shock raised deaths of despair more where S_c was higher. The share of the gradient a feature accounts for is the shift in S_c that the Great Migration produces between the 25th and 75th percentiles of predicted migration (Figure 6), times β_3, divided by the gradient estimated on the same sample with the same controls but without the interaction.[1]

The features are themselves outcomes of the Great Migration and are correlated with one another, so the shares are not additive. A feature that accounts for part of the gradient is a channel the evidence is consistent with, not an estimate of mediation.

6.2 Divided polities and public goods

If segregation divided local electorates and weakened support for shared services (Trounstine, 2016), destinations should show weaker support for redistribution and weaker public provision. Weaker support for redistribution could also lead to a reallocation across the local governments within a commuting zone. These effects could even move in opposite directions, for instance with a reduction in school spending in urban districts and an increase in suburban ones (Derenoncourt, 2022).

Between the 25th and 75th percentiles of predicted migration, the Great Migration raised racial segregation by 0.71 standard deviations and White residents' opposition to redistribu-

[1]The gradient falls by more than this share for some features when the interaction is added (Appendix Table A16). This is because that reduction attributes to the characteristic all of its correlation with predicted migration, including correlation that does not come from the Great Migration, for instance if both are high in large, older industrial metropolitan areas. The share, on the other hand, only counts the part that the Great Migration produced. The intervals reflect only the sampling error in β_3. First stages are computed as in Table 2 (Borusyak et al., 2022).

tion by 0.65 standard deviations (Figure 6). Opposition is measured in the 2014–2020 waves of the Cooperative Congressional Election Study (CCES), with an index that averages standardized opposition to welfare, health-care, and education spending and to a minimum-wage increase. The China shock raised mortality more where White residents opposed redistributive spending more; the interaction is significant for Black men and marginally significant for White men. The Great Migration's effect on opposition accounts for 15 percent of the White accidental-poisoning gradient (95 percent interval: −0.2 to 30) and 18 percent of the Black gradient (6 to 29).

Using the local government finance data of Feler and Senses (2017), I estimate the same specification with the 2000–2007 change in local spending per resident as the outcome (Appendix Table A17). The responses of total, education, public-safety, and welfare spending to the shock did not differ across destinations, but the response of health and hospital spending did. A one-standard-deviation shock raised local health and hospital spending by about $74 per resident over a decade in commuting zones at the 25th percentile of predicted migration and left it unchanged at the 75th percentile: a gradient of −$79 per resident (SE = 36), against average spending of about $320. In the decade before the shock, the same gradient had the opposite sign and was not significant.

The shock also held back health spending where White residents opposed redistribution more, by $31 per resident for each standard deviation of opposition (SE = 7), and the Great Migration's effect on opposition accounts for about 29 percent of the health-spending gradient (95 percent interval: 17 to 41). High-migration destinations, in other words, did not expand health services when the shock hit.

Finally, I find no evidence of greater local government fragmentation, and the association with city-manager government in 1996 (Grumbach et al., 2026) is imprecise (0.29 standard deviations, $p = 0.11$). These findings fit Trounstine (2016)'s account, but I measure attitudes using CCES waves from after the China shock. Trade exposure itself changed local politics (Autor et al., 2020). So part of the association may reflect the shock changing attitudes, rather than those attitudes shaping the mortality response.

6.3 Segregation and concentrated poverty

In Massey (1990)'s account, racial segregation confines rising Black poverty to Black neighborhoods, so a loss of income produces neighborhoods of concentrated poverty. The hypothesis predicts that the Great Migration raised racial residential segregation and concentrated poverty, and that the China shock was deadlier where poverty was more concentrated. I am not testing exactly Massey's hypothesis, where residential segregation would itself have to cause poverty. Rather, I am asking whether the Great Migration increased residential segregation and concentrated poverty, which then made these places more vulnerable when the manufacturing shock arrived.

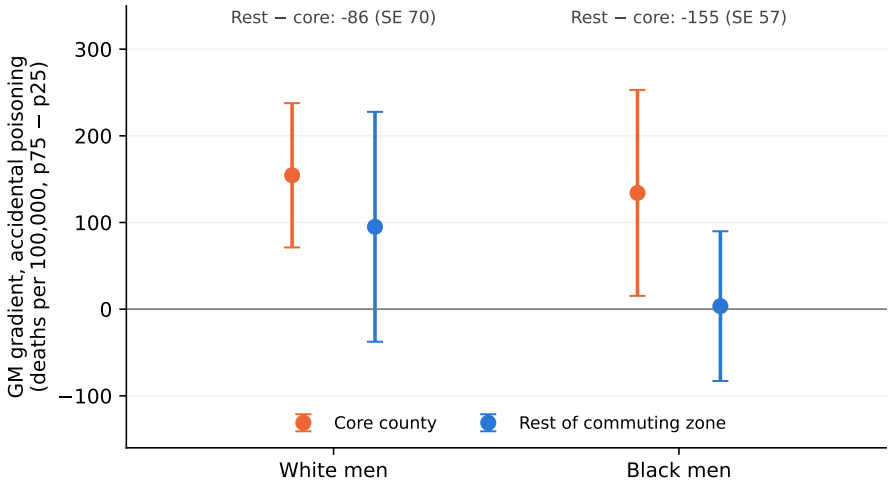
I find that the first link holds: the Great Migration increased racial segregation, the share of poor residents in tracts with at least 40 percent poverty (0.72 standard deviations), the corresponding share for poor Black residents (0.81), and class segregation among all residents (0.62; Figure 6).

The second link holds for Black men's accidental drug poisoning. The shock raised deaths more where poverty was more concentrated, and the Great Migration's effect on concentrated poverty accounts for 36 percent of the Black gradient (95 percent interval: 9 to 63). The share is similar for related measures: 42 percent (11 to 72) for the concentration of poor Black residents, 45 percent for class segregation among all residents, and 42 percent for racial segregation. These measures describe one channel, observed in several ways, which accounts for roughly 35 to 45 percent of the Black accidental-poisoning gradient. The channel does not account for the Black broader-outcome gradient (shares of -1 to 17 percent, none significant), and it does not operate for White men: concentrated poverty accounts for 9 percent of the White accidental-poisoning gradient and concentrated White poverty for 1 percent, and neither is significant.

The geography fits. Comparing each commuting zone's urban-core county with the rest of the zone, the Black accidental-poisoning gradient is 134 deaths per 100,000 in the core and 4 elsewhere, a significant difference, while the White gradient is large in both (155 and 95; the difference is not significant; Figure 7 and Appendix Table A18). As Massey (1990) predicts, the channel is specific to Black men. It cannot explain why White men in the same

destinations responded as strongly.

Figure 7: Black deaths rose in urban cores; White deaths rose across the commuting zone, 2000–2013



Notes: This figure plots the Great Migration gradient in the effect of the China shock on accidental drug poisoning, estimated separately for each CZ's urban-core county and for the rest of the CZ. The gradient is the difference in the effect of a one-standard-deviation China shock on cumulative 2000–2013 deaths per 100,000 men ages 20–39 between the 75th and 25th percentiles of predicted Great Migration rank. The core is the county with the most principal-city tracts (2010 definitions), plus any county lying entirely within a principal city. Estimates use the specification of Table 2; bars are 95 percent confidence intervals based on exposure-robust standard errors. The difference between the rest of the CZ and the core is estimated directly and reported above each pair; Appendix Table A18 reports the estimates.
Sources: NCHS restricted-use mortality files, SEER county populations, LTDB 2010 principal-city tract flags.

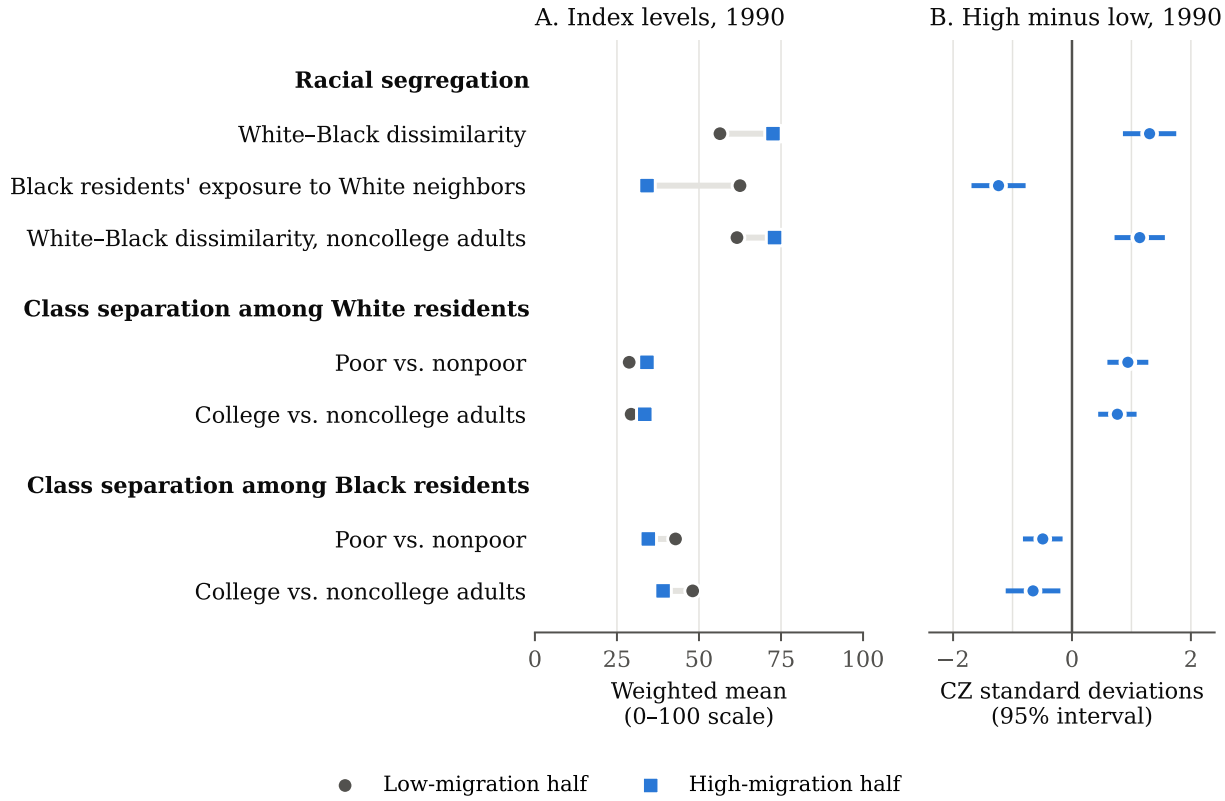
6.4 Isolation and the exit of the middle class

According to Wilson (1991)', the departure of middle-class residents particularly form poor neighborhoods, leads to isolation of those left behind: from employed neighbors, social networks, and role-model formation. While this is conceptualized as an outcome of manufacturing decline in the 1970s, the same prediction should apply to declining cities as the relatively well-off exit them. Applying this to the Great Migration would mean predicting greater sorting by class within the racial groups, predicting that poor residents becoming more isolated from non-poor neighbors, thereby leading to a deadlier shoock in places with higher isolation. This framework could work for both the Black and the White "underclass".

For the Black residents, this is not the case. While predicted migration increased the isolation of poor Black residents from all non-poor neighbors (0.83 standard deviations), it

reduced both class and education based segregation among Black residents (by 0.59 and 0.55 standard deviations; Figure 6). Looking at the levels in 1990 shows us that the poor-nonpoor dissimilarity among Black residents was 34.6 in high-migration destinations and 42.9 in low-migration destinations. The college–noncollege dissimilarity was 39.1 and 48.1, while both indices were higher among White residents of high-migration destinations (Figure 8). This is a strong indicator of sorting based on racial lines. The isolation of poor Black residents from all residents accounts for 40 percent of the Black accidental poisoning gradient, as much as the other measure for concentrated poverty. Thus, the evidence does not clearly line up with Wilson's mechanism.

Figure 8: Racial segregation and class separation within race in low- and high-migration destinations, 1990



Notes: Low- and high-migration halves are the lower and upper halves of predicted Great Migration rank among the 130 commuting zones in the mortality sample. Panel A reports commuting-zone means weighted by 2000 population, on a 0–100 scale; Panel B divides the high-minus-low difference by the measure's commuting-zone standard deviation and shows 95 percent confidence intervals based on standard errors clustered by state. All measures are for 1990 and use census tracts mapped to commuting zones with the complete county crosswalk. Dissimilarity indices compare White and Black residents, poor and nonpoor residents, or adults with and without a four-year college degree; class separation within race compares residents of the same race. Black residents' exposure to White neighbors is the White population share of the tract of the average Black resident, so higher values mean more contact. The college comparison among Black residents covers 128 commuting zones. The figure describes differences across destinations, not causal effects.
Sources: Longitudinal Tract Database; NHGIS 1990 census tract tables; Derenoncourt (2022).

For White residents, the Great Migration raised class segregation (0.64 standard deviations) and education segregation (0.52) among White residents, as well as the isolation of poor White residents (0.56). I find that the shock was significantly less deadly for White men where poor White residents, or poor Black residents, were more isolated from nonpoor neighbors. Here, education segregation is the exception. it accounts for 11 percent of the White gradient, and 19 percent of the Black gradient. While each of these estimates is im-

precise, together they suggest that the sorting of college-educated and non-college residents into separate neighborhoods explains a modest part of the gradient, somewhere between a tenth and a fifth.

6.5 Other features of the destinations

Three other characteristics could have made the shock deadlier in high-migration destinations without operating through segregation.

Incarceration. The Great Migration raised incarceration (Derenoncourt, 2022): the 1983–2000 prison rate is 0.53 standard deviations higher at the 75th than at the 25th percentile of predicted migration. However, the China shock was not significantly deadlier in places with higher incarceration.

Routine work. Adda and Fawaz (2020) show that the Chinese import competition was harmful to health where the jobs were more routine. While their general result holds here for Black men. However, the routine intensity of manufacturing work is unrelated to predicted migration, and t accounts for none of the gradient. While routine work made places more vulnerable to the shock, but high-migration destinations did not have much more of it.

Drug availability. State opioid policy shapes how an economic shock translates into overdose deaths. Controlling for the number of opioid-restricting state laws in force from 2006 to 2012, as in Pierce and Schott (2020), lowers the gradient for accidental drug poisoning from 105 to 65 deaths per 100,000 for White men and from 82 to 51 for Black men, and for the broader outcome from 69 to 19 and from 98 to 56; the estimates are imprecise. However, opioid-law exposure is only weakly related to predicted migration (0.32 standard deviations between the 25th and 75th percentiles, $p = 0.17$, 32 states). So, the decline reflects an overlap between the state policy environment and the migration geography rather than a channel of the Great Migration. Because laws vary only across states, this comparison has little independent variation.

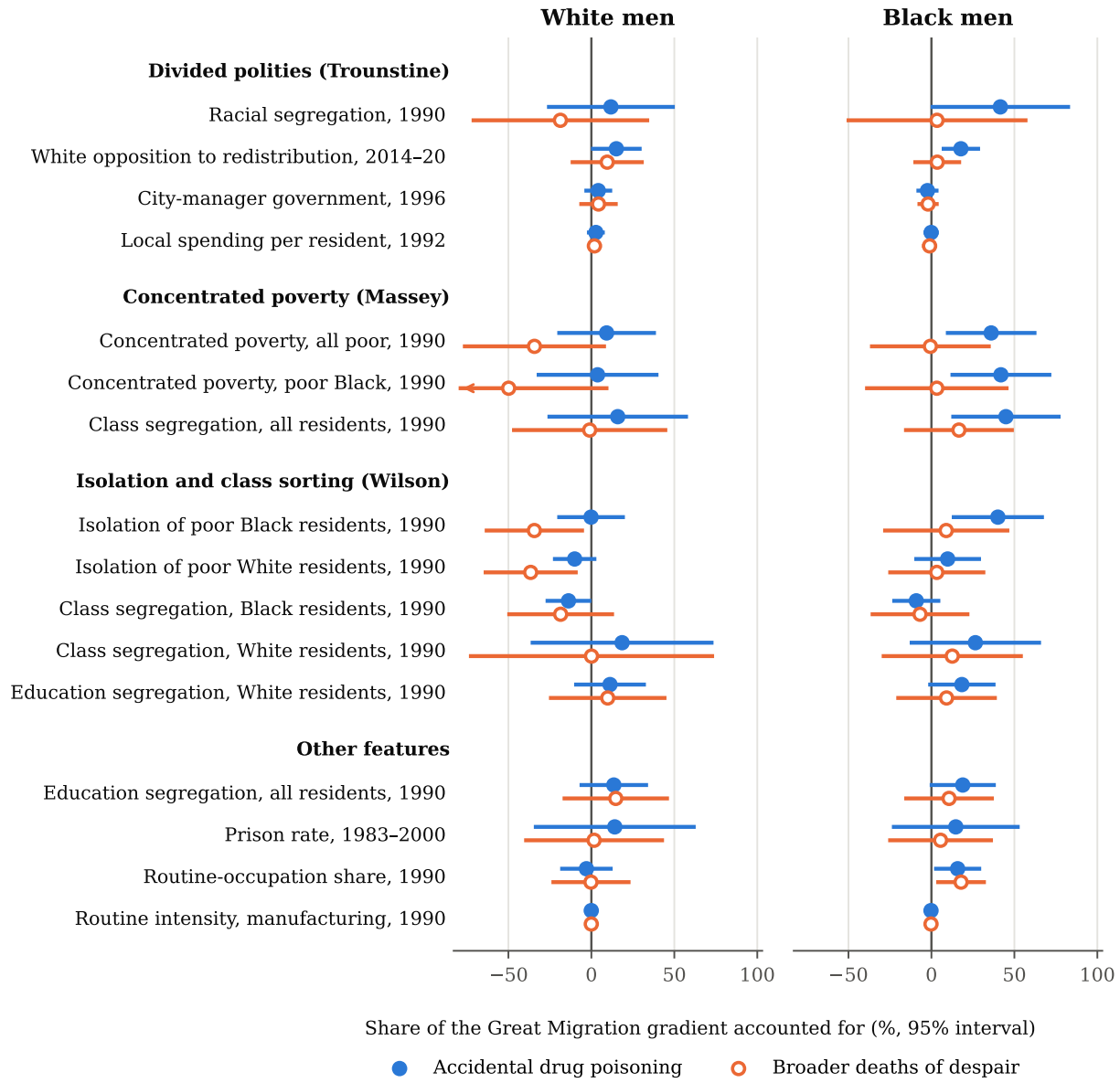
6.6 Which channels operate, for whom, and how much

The results are summarized in Figure 9 and Table 6. Two conclusions follow.

First, for Black men the Great Migration gradient in accidental drug poisoning works through concentrated poverty, as Massey (1990) predicts. Racial segregation, class segregation, concentrated poverty, and the isolation of poor Black residents each explain about 35 to 45 percent of it. These are overlapping measures of the same channel. About a fifth of the Black gradient is associated with White opposition to redistribution and a sixth with routine work. None of these channels accounts for a significant part of the Black broader-outcome gradient, except routine work (18 percent).

For White men the various measures of segregation in the destinations do not account for much of the gradient. Concentrated poverty, class segregation, isolation, incarceration, fiscal capacity, local institutions, and occupational structure each explain very little of the gradient. White opposition to redistribution explains 15 percent of the accidental-poisoning gradient, and education segregation explains a small component of it. Thus, for the White gradient, most of the available evidence supports Trounstine (2016)'s argument, along with the health-expenditure results. While the mechanisms around Black responses align with the segregation literature, my results explain one-third of the White gradient.

Figure 9: Which destination features account for the Great Migration gradient, and for whom



Notes: Each point is the share of the Great Migration gradient in the effect of the China shock accounted for by one destination feature, estimated from a separate regression that adds the China shock times the feature (instrumented by the other-country shock times the feature) to the specification of Table 2. The share is the shift in the feature produced by moving from the 25th to the 75th percentile of predicted migration (Figure 6) times the shock-by-feature coefficient, divided by the gradient on the same sample with the same controls but without the shock-by-feature term. Lines are 95 percent intervals from the exposure-robust standard error of the shock-by-feature coefficient; arrows mark intervals that extend beyond the plotted range. Filled markers: accidental drug poisoning; hollow markers: suicide, drug poisoning, and alcoholic liver disease combined. Shares of different features overlap and are not additive. Each label gives the year in which the feature is measured; mortality outcomes are cumulative 2000–2013 deaths per 100,000 men ages 20–39. Appendix Table A16 reports the underlying estimates.

7 Conclusion

The mortality toll of the China shock depended on where it fell. In destinations shaped by the Great Migration, the same import shock, with no larger manufacturing job losses, raised accidental drug-poisoning deaths among young men far more than elsewhere. Moving from the 25th to the 75th percentile of predicted Great Migration increases the effects of a one-standard-deviation China shock: about 105 deaths per 100,000 White men and 82 per 100,000 Black men. These gradients cannot be distinguished statistically. This is not driven due to high migration destinations being more exposed to import competition, loss of more manufacturing jobs, or outmigration. This pattern is in the same direction for all and White men with the PNTR design of Pierce and Schott (2016). Thus, the deaths of despair were not just a White phenomenon, and the most exposed places were shaped by their racial history than by the magnitude of the economic shock.

However, despite similar effects, I also find that these channels differed across race: while Black deaths of despair were concentrated in the urban centers, associated with concentrated poverty and isolation that seems to be closely associated to racial segregation instead of Black middle-class exodus. This channel explains between 35 to 45 percent of the gradient. Among White men, the excess mortality happened across the whole commuting zone, and neighborhood poverty, class sorting, incarceration, or local fiscal and institutional differences do not explain most of the gradient. White residents are found to oppose redistributive spending, and there is separation of college and non-college residents, both of which modestly explain the White patterns.

My findings demonstrate that we cannot understand the costs of deindustrialization from the job losses. The same shock was more lethal in places where history had concentrated poverty, divided local politics, and made public provision less responsive, and this history made both White and Black residents more vulnerable. Thus, policies which respond to trade and other local labor market shocks by targeting the places with the largest job losses are likely to miss the places where similar losses are more deadly.

My design also has its limits. Predicted Great Migration identifies places with a particular history, but it does not separate the causal effect of the Great Migration from the institutions

and economic structures that came along with it. Also, these destination features that I study are correlated outcomes of that history, and are not separately identified mediators. For the education results, I rely on 1990 population shares and on how I allocate deaths with unknown education. I am also unable to provide a comprehensive explanation for the White mortality gradient.

Further work could examine drug markets, treatment capacity, state drug policy, and public services, as well as how these destinations responded to later shocks.

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Tables

Table 1: Great Migration destinations: characteristics and mortality

		Great Migration sample			
	All CZs	Full sample	Bottom quartile	Top quartile	Top − bottom
Commuting zones	722	130	36	33	–
Panel A. Commuting zones, 2000					
Predicted Great Migration rank	–	62.6	9.6	88.9	79.3
China import shock, 2000–2012	0.89	0.89	0.82	0.72	−0.09
Manufacturing share of employment	16.2	16.0	15.5	14.7	−0.8
Black share of population	12.4	10.5	4.8	17.1	12.4
College-educated share of population	53.5	55.9	58.8	55.3	−3.5
Foreign-born share of population	14.7	17.5	16.6	16.3	−0.3
Routine-occupation share of employment	31.9	32.9	31.7	33.7	2.0
Panel B. Mortality, men ages 20–39 (cumulative deaths per 100,000)					
All causes, 1990–1999	–	1893.9	1786.1	2117.2	331.1
Accidental drug poisoning, White men	–	262.0	232.3	310.2	77.9
Accidental drug poisoning, Black men	–	149.0	162.4	166.6	4.2
Broader outcome, White men	–	612.3	629.5	655.1	25.6
Broader outcome, Black men	–	403.9	441.4	428.7	−12.7
Panel C. Neighborhoods, 1990 (percent)					
White–Black dissimilarity	–	67.4	55.1	77.0	21.8
Poor residents in 40%+ poverty tracts	–	13.3	7.0	20.6	13.6
Poor Black residents in 40%+ poverty tracts	–	24.6	14.6	34.9	20.3

Notes: Entries are means weighted by 2000 commuting-zone population. "All CZs" refers to 722 commuting zones; the Great Migration sample is the 130 non-Southern destinations used in the mortality analysis. The bottom and top quartiles of predicted Great Migration rank contain 36 and 33 commuting zones. Here, the China import shock is the 2000–2012 change in Chinese import penetration in the United States, in percentage points, expressed as a ten-year equivalent (Autor et al., 2021); it is the exposure measure used throughout the paper. Shares are in percent and measured in 2000. The broader outcome combines suicide, alcoholic liver disease, and accidental or undetermined-intent drug poisoning. Panel C uses 1990 tract data: dissimilarity is on a 0–100 scale, and the poverty rows give the percent of poor residents who live in tracts with poverty rates of 40 percent or more (124 commuting zones have Black poverty data). Mortality and neighborhood measures are shown only for the Great Migration sample.
Sources: NCHS restricted-use mortality files; SEER population estimates; Autor et al. (2021); Derenoncourt (2022); Longitudinal Tract Database.

Table 2: Import competition and mortality across Great Migration destinations, 2000–2013

	All men		White men		Black men	
	No interaction (1)	Interacted (2)	No interaction (3)	Interacted (4)	No interaction (5)	Interacted (6)
Panel A. Accidental drug poisoning						
China import shock (one SD)	17.7	23.6	22.2	28.6	9.2	11.3
	(18.2)	(17.0)	(20.1)	(19.8)	(22.0)	(17.7)
China shock × Predicted	–	98.1***	–	105.4***	–	81.7**
migration (one IQR)		(37.4)		(38.8)		(41.0)
Panel B. Suicide, alcoholic liver disease, and drug poisoning						
China import shock (one SD)	31.2*	31.3*	38.5*	38.3*	33.7	33.4
	(17.7)	(18.3)	(19.9)	(21.9)	(28.6)	(22.8)
China shock × Predicted	–	68.3*	–	69.3*	–	97.7*
migration (one IQR)		(35.9)		(37.9)		(54.4)
First-stage F: China shock	88.11	46.90	88.11	46.90	88.11	46.90
First-stage F: interaction	–	52.21	–	52.21	–	52.21
SW conditional F: shock	–	38.48	–	38.48	–	38.48
SW conditional F: interaction	–	37.97	–	37.97	–	37.97
Commuting zones	130	130	130	130	130	130
Industry clusters	136	136	136	136	136	136
2000 population weights	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports the effects of a one-standard-deviation increase in Chinese import exposure on cumulative mortality per 100,000 men ages 20–39 over 2000–2013. Cumulative mortality is the sum of annual death rates. The two outcomes are accidental drug poisoning; and a broader outcome that adds suicide, alcoholic liver disease, and drug poisoning of undetermined intent. Columns 1, 3, and 5 estimate the average response. Columns 2, 4, and 6 allow it to vary with predicted Great Migration rank. In these columns, the China-shock coefficient is the effect at the median Great Migration rank (50.5); the interaction gives the change between the 25th and 75th percentiles (57 rank points). All regressions include the Autor–Dorn–Hanson (ADH) covariates: the 2000 manufacturing employment share, college-educated and foreign-born population shares, female employment share, routine-intensive employment share, offshorability measure, population shares ages 0–17, 40–64, and 65 and older, and Census division fixed effects (Autor et al., 2021). They also control for the sum of lagged industry exposure shares and cumulative 1990–1999 all-cause mortality among men ages 20–39. Columns 2, 4, and 6 additionally include predicted Great Migration rank, its interactions with the manufacturing employment share and the exposure-share sum, and the share of the 1940 urban population consisting of Black migrants from the South during 1935–1940 (Derenoncourt, 2022). These additional controls are omitted from columns 1, 3, and 5. All regressions are weighted by 2000 CZ population. Standard errors for the China-shock terms use exposure-robust inference clustered by three-digit industry. First-stage F statistics follow Borusyak et al. (2022): each comes from the industry-level IV regression of the aggregated treatment on the aggregated shift-share instruments, instrumented by the industry shocks, with standard errors clustered by three-digit industry. For the interacted model, the table reports this statistic for the China shock and for its interaction with predicted migration, and the corresponding Sanderson–Windmeijer conditional F statistics. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Mortality responses across destinations, 2000–2013: a Black–White comparison

	Migration gradient (75th − 25th percentile)
Panel A. Accidental drug poisoning	
Black men	81.7**
	(41.0)
White men	105.4***
	(38.8)
Black − White difference	−23.7
	(28.6)
Joint-test p-value	0.407
Panel B. Suicide, alcoholic liver disease, and drug poisoning	
Black men	97.7*
	(54.4)
White men	69.3*
	(37.9)
Black − White difference	28.4
	(39.6)
Joint-test p-value	0.472

Notes: The table compares the change in the mortality effect between the 25th and 75th percentiles of predicted Great Migration rank for Black and White men. Effects are for a one-standard-deviation China shock and are measured in cumulative deaths per 100,000 men ages 20–39 over 2000–2013. The narrow outcome is accidental drug poisoning. The broader outcome adds suicide, alcoholic liver disease, and drug poisoning of undetermined intent. The sample, controls, and 2000 population weights follow Table 2. The two equations are estimated jointly, allowing for covariance between the race-specific estimates. The reported test concerns equality of the two gradients. Exposure-robust standard errors in parentheses are clustered by three-digit industry. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Import competition and employment across Great Migration destinations, 2001–2012

	Manufacturing		Nonmanufacturing	
	No interaction	Interacted	No interaction	Interacted
Change in employment relative to working-age population, 2001–2012 (percentage points)				
China import shock (one SD)	−0.567***	−0.618***	0.030	0.015
	(0.155)	(0.155)	(0.646)	(0.643)
China shock × Predicted migration (one IQR)	–	0.523	–	1.217
		(0.505)		(1.027)
Commuting zones	130	130	130	130
Industry clusters	136	136	136	136

Notes: The table reports the effects of Chinese import competition on manufacturing and nonmanufacturing employment relative to the working-age population over 2001–2012. Effects are in percentage points for a one-standard-deviation China shock (0.454). The sample contains the same 130 commuting zones as the mortality analysis. In the interacted columns, the China-shock coefficient is evaluated at the median predicted Great Migration rank (50.5), and the interaction gives the change between the 25th and 75th percentiles (ranks 18 and 75). Controls include lagged manufacturing employment shares and the sum of industry exposure shares; the interacted columns also interact these controls with predicted migration. Exposure-robust standard errors in parentheses are clustered by three-digit industry. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Education differences in mortality responses at high predicted migration, 2000–2013

Group	Difference	Standard error	p-value
Panel A. Accidental drug poisoning			
All men	78.5**	32.8	0.017
White men	106.0***	39.1	0.007
Black men	72.2**	28.6	0.011
Panel B. Suicide, alcoholic liver disease, and drug poisoning			
All men	95.5**	38.4	0.013
White men	122.2***	46.5	0.009
Black men	76.0**	31.1	0.014

Notes: The table compares the effect of import competition among men with high school or less with the effect among men with education beyond high school. Entries subtract the latter from the former at the 75th percentile of predicted Great Migration rank. Effects are for a one-standard-deviation China shock, in cumulative deaths per 100,000 men ages 20–39 over 2000–2013. The narrow outcome is accidental drug poisoning. The broader outcome adds suicide, alcoholic liver disease, and drug poisoning of undetermined intent. All-men and White-male estimates use 130 commuting zones; Black-male estimates use 129. All regressions include the ADH covariates (the 2000 manufacturing employment share, college-educated and foreign-born population shares, female employment share, routine-intensive employment share, offshorability measure, population shares ages 0–17, 40–64, and 65 and older, and Census division fixed effects), predicted Great Migration rank, the share of the 1940 urban population consisting of Black southern migrants who arrived during 1935–1940, the sum of lagged industry exposure shares, and interactions of predicted migration with the manufacturing share and the exposure-share sum. They also include education-specific lagged all-cause mortality and use 2000 population weights. Deaths with unknown education are allocated in proportion to known-education deaths; fixed 1990 education shares divide the annual population denominators. Standard errors allow for covariance between the education-specific equations and use exposure-robust inference clustered by three-digit industry. Positive differences indicate larger effects among men with high school or less. Stars use unrounded p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Share of the Great Migration gradient accounted for by each destination condition

	White men		Black men	
Destination condition	Accidental	Broader	Accidental	Broader
H1. Divided polities and public goods (Trounstine)				
Racial segregation (White–Black Theil)	12% [-27, 50]	-19% [-72, 35]	42% [0, 84]	3% [-51, 58]
White opposition to redistribution (CCES, 2014–20)	15% [0, 30]	10% [-12, 32]	**18% [6, 29]**	4% [-11, 18]
City-manager government, 1996	4% [-4, 13]	4% [-7, 16]	-2% [-9, 4]	-2% [-8, 4]
Local governments per capita, 1992 (log)	0% [0, 0]	0% [0, 0]	0% [0, 0]	0% [0, 0]
Local spending per resident, 1992 (log)	3% [-3, 8]	2% [-3, 6]	0% [-5, 4]	-1% [-6, 3]
H2. Segregation concentrates poverty (Massey)				
Poor in 40%+ poverty tracts, all	9% [-21, 39]	-34% [-77, 9]	**36% [9, 63]**	-1% [-37, 36]
Poor in 40%+ poverty tracts, Black	4% [-33, 40]	-50% [-110, 10]	**42% [11, 72]**	3% [-40, 46]
Poor in 40%+ poverty tracts, White	1% [-15, 17]	-18% [-42, 6]	7% [-20, 34]	2% [-33, 36]
Class segregation, all residents (Theil)	16% [-26, 58]	-1% [-48, 46]	**45% [12, 78]**	17% [-17, 50]
H3. Isolation and middle-class exit (Wilson)				
Isolation of poor Black residents	0% [-21, 20]	**-34% [-64, -4]**	**40% [12, 68]**	9% [-29, 47]
Isolation of poor White residents	-10% [-23, 3]	**-37% [-65, -8]**	10% [-10, 30]	3% [-26, 32]
Class segregation, Black (Theil)	-14% [-28, 0]	-19% [-51, 14]	-9% [-24, 5]	-7% [-37, 23]
Education segregation, Black (Theil)	-2% [-20, 15]	3% [-24, 31]	-8% [-22, 7]	-10% [-38, 19]
Class segregation, White (Theil)	19% [-37, 74]	0% [-74, 74]	26% [-13, 66]	13% [-30, 55]
Education segregation, White (Theil)	11% [-10, 33]	10% [-26, 45]	18% [-2, 39]	9% [-21, 39]
Other destination conditions				
Education segregation, all residents (Theil)	14% [-7, 34]	15% [-17, 47]	19% [-1, 39]	11% [-16, 38]
Prison rate, 1983–2000	14% [-35, 63]	2% [-41, 44]	15% [-24, 53]	6% [-26, 37]
Routine-occupation share	-3% [-19, 13]	0% [-24, 24]	**16% [2, 30]**	**18% [3, 33]**
Routine task intensity, manufacturing	0% [0, 0]	0% [0, 0]	0% [0, 0]	0% [0, 0]

Notes: Each cell is from a separate regression: the Table 2 specification plus the China shock times the standardized condition, instrumented by the other-country shock times the condition, with the condition and its interactions with the manufacturing share and the share sum as controls. All mortality regressions include the ADH covariates (the 2000 manufacturing employment share, college-educated and foreign-born population shares, female employment share, routine-intensive employment share, offshorability measure, population shares ages 0–17, 40–64, and 65 and older, and Census division fixed effects), predicted Great Migration rank, the share of the 1940 urban population consisting of Black southern migrants who arrived during 1935–1940, the sum of lagged industry exposure shares, and interactions of predicted migration with the manufacturing share and the exposure-share sum. They also include cumulative 1990–1999 all-men all-cause mortality and use 2000 CZ population weights. The regression of each condition on predicted migration includes Derenoncourt's 1940 controls: educational upward mobility, the manufacturing share of the labor force, the 1935–1940 Black southern-migrant share of the urban population, and Census region fixed effects. The share is the shift in the condition produced by the Great Migration (Figure 6) times the shock-by-condition coefficient, divided by the gradient on the same sample with the same controls but no shock-by-condition term. 95% intervals from the exposure-robust standard error of the shock-by-condition coefficient (Borusyak et al., 2022); bold when the interval excludes zero. Conditions are measured in 1990 unless the row gives another year; mortality outcomes are cumulative 2000–2013 deaths per 100,000 men ages 20–39. Coefficients, gradients, and first-stage statistics are in Table A16.

Appendix Tables

Table A1: Definitions of the two mortality outcomes

Measure	ICD-10 codes	Key exclusions
Accidental drug poisoning (narrow)	X40–X44.	Excludes suicide, undetermined-intent poisoning, and non-drug poisoning.
Broader mortality	Suicide, all methods (U03, X60–X84, Y87.0). Alcoholic liver disease (K70). Fatal drug overdose (X40–X44, Y10–Y14): accidental or undetermined-intent drug poisoning.	Excludes accidental poisonings involving non-drug substances. Suicide overdoses enter once through the suicide component.

Notes: Both outcomes are constructed from underlying causes of death in the restricted-use National Center for Health Statistics mortality records. Accidental drug poisoning is a subset of the broader outcome. Suicide by drug poisoning enters the broader measure once, through its suicide component. The cumulative analyses sum annual rates per 100,000 people ages 20–39 over 2000–2013; the trade-policy event study uses annual rates for the same causes. Historical all-cause mortality enters separately as a control. ICD-10 denotes the International Classification of Diseases, Tenth Revision. Appendix Table A4 separately retains the Autor–Dorn–Hanson poisoning benchmark and its original outcome definition.

Table A2: The Great Migration gradient by cause of death, 2000–2013

	Effect at p75 of predicted GM			Gradient, p75 − p25		
Cause	All	White	Black	All	White	Black
Broader outcome (main estimates)	60.7**	68.1**	75.4**	68.3*	69.3*	97.7*
	(27.2)	(30.8)	(38.2)	(35.9)	(37.9)	(54.4)
Drug overdose (accidental and undetermined)	64.7**	73.5**	51.2	75.1**	80.8**	70.1
	(27.3)	(30.0)	(35.5)	(35.1)	(36.9)	(43.3)
Accidental drug poisoning	65.8**	73.9**	46.4	98.1***	105.4***	81.7**
	(27.9)	(30.1)	(33.6)	(37.4)	(38.8)	(41.0)
Undetermined-intent drug poisoning	-1.1	-0.4	4.7	-23.0	-24.5	-11.6
	(11.6)	(13.3)	(10.0)	(17.7)	(20.5)	(16.2)
Suicide	-4.3	-5.7	22.7*	-7.8	-12.1	25.4
	(8.8)	(9.1)	(11.9)	(12.0)	(12.6)	(25.7)
Alcoholic liver disease	0.2	0.3	1.5	1.0	0.5	2.2
	(1.7)	(2.0)	(2.1)	(2.2)	(2.8)	(2.7)

Notes: This table reports the effect of the China shock on mortality from each cause in high-migration destinations and its Great Migration gradient. The unit of observation is a CZ. The dependent variable is the cumulative 2000–2013 death rate per 100,000 men ages 20–39 from the listed cause. The first three columns report the effect of a one-standard-deviation China shock at the 75th percentile of predicted Great Migration rank; the last three report the difference between the 75th and 25th percentiles. Drug overdose combines accidental and undetermined-intent drug poisoning; the broader outcome adds suicide and alcoholic liver disease. Each row re-estimates the specification of Table 2. The instrument for the China shock is the change in Chinese imports to other high-income countries, applied to lagged CZ industry shares; each interaction term is instrumented by the instrument times the corresponding variable. All specifications include the controls of Table 2, including cumulative 1990–1999 all-cause mortality, and are weighted by CZ population in 2000. Exposure-robust standard errors, clustered by three-digit industry following Borusyak et al. (2022), are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Sources: NCHS restricted-use mortality files, Autor et al. (2021), Derenoncourt (2022).

Table A3: Mortality effects at selected predicted migration ranks, 2000–2013

	All men	White men	Black men
Panel A. Accidental drug poisoning			
Effect at 25th-percentile GM	−32.3	−31.5	−35.2**
	(20.9)	(23.8)	(13.8)
Effect at median GM	23.6	28.6	11.3
	(17.0)	(19.8)	(17.7)
Effect at 75th-percentile GM	65.8**	73.9**	46.4
	(27.9)	(30.1)	(33.6)
75th-minus-25th-percentile gradient	98.1***	105.4***	81.7**
	(37.4)	(38.8)	(41.0)
Observed cumulative mortality mean	230.1	262.0	149.0
75th-percentile effect as share of mean	28.6%	28.2%	31.1%
Panel B. Suicide, alcoholic liver disease, and drug poisoning			
Effect at 25th-percentile GM	−7.6	−1.2	−22.3
	(23.1)	(26.1)	(30.8)
Effect at median GM	31.3*	38.3*	33.4
	(18.3)	(21.9)	(22.8)
Effect at 75th-percentile GM	60.7**	68.1**	75.4**
	(27.2)	(30.8)	(38.2)
75th-minus-25th-percentile gradient	68.3*	69.3*	97.7*
	(35.9)	(37.9)	(54.4)
Observed cumulative mortality mean	557.8	612.3	403.9
75th-percentile effect as share of mean	10.9%	11.1%	18.7%

Notes: Entries give the effect of a one-standard-deviation China shock at the 25th, 50th, and 75th percentiles of predicted Great Migration rank (18, 50.5, and 75), and the difference between the 75th- and 25th-percentile effects. Estimates come from the interacted models in Table 2. Outcomes are cumulative deaths per 100,000 men ages 20–39 over 2000–2013. The narrow outcome is accidental drug poisoning. The broader outcome adds suicide, alcoholic liver disease, and drug poisoning of undetermined intent. The last two rows compare the effect at the 75th percentile with the outcome mean weighted by 2000 population. Exposure-robust standard errors in parentheses are clustered by three-digit industry. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Comparison with the Autor–Dorn–Hanson poisoning outcome

	Male poisoning mortality	
	No interaction (1)	Interacted (2)
China import shock	18.0***	18.9**
(one SD)	(6.5)	(8.1)
China shock × Predicted migration	–	70.6***
(one IQR)		(17.7)
Predicted migration main effect	–	−82.8***
(one IQR)		(31.4)
Effect at p75 of Predicted migration	–	49.3***
(one SD)		(9.5)
K–P rk Wald F	73.48	13.34
SW conditional F: shock	–	46.09
SW conditional F: interaction	–	34.49
Commuting zones	130	130
State clusters	32	32
Population weights	Yes	Yes

Notes: This table uses the poisoning outcome of Autor et al. (2019) for men ages 20–39 in the post-2000 period, expressed as decadal deaths per 100,000. It is a separate benchmark, with a different poisoning definition from the two principal outcomes. The sample contains the 130 Great Migration commuting zones. Effects are scaled by the standard deviation of the Autor–Dorn–Hanson import-exposure measure (0.622). In column 2, the China-shock coefficient is evaluated at the median predicted migration rank (50.5); the interaction gives the 75th-minus-25th-percentile difference (57 rank points). The last effect row evaluates the response at rank 75. Chinese import growth in other high-income countries instruments trade exposure and its interaction. Regressions use the Autor–Dorn–Hanson controls and population weights; the interacted model also includes predicted migration and its interaction with initial manufacturing employment. Standard errors are clustered by state. The percentile-specific estimates are calculated here. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Correlation between predicted Great Migration and Chinese import exposure, 2000–2012

Adjustment	Correlation (r)	Squared correlation (r^2)	Commuting zones
Panel A. Population weights			
No controls (raw correlation)	-0.138	0.019	130
Baseline controls	0.127	0.016	130
Baseline controls + migration interactions	0.090	0.008	130
Panel B. Equal weight for each commuting zone			
No controls (raw correlation)	-0.086	0.007	130
Baseline controls	0.045	0.002	130
Baseline controls + migration interactions	-0.023	0.001	130

Notes: The table reports correlations across the 130 commuting zones in the mortality sample. Import exposure is the observed 2000–2012 increase in Chinese import penetration, expressed as a ten-year equivalent. Adjusted correlations remove the same controls from both import exposure and predicted migration, using the indicated weights. The additive controls are initial manufacturing employment; Census divisions; college, foreign-born, female-employment, routine-employment, offshorability, and age measures; the early Black migrant share; the sum of industry exposure shares; and lagged all-men all-cause mortality. The additional-interaction rows also remove interactions of predicted migration with manufacturing employment and the sum of exposure shares. The squared correlation is the share of residual variation associated with the other residualized measure. These are descriptive associations.

Table A6: Employment and wage responses by race in Great Migration destinations

Outcome	White shock effect	Black shock effect	Hispanic shock effect	Black–White	Hispanic–White	Min. SW F
Panel A. Change from 2000 to 2010						
Manufacturing	−0.142***	−0.200***	−0.090**	−0.058	0.052	25.27
employment	(0.036)	(0.077)	(0.043)	(0.075)	(0.044)	
Nonmanufacturing	0.009	0.048*	0.003	0.040	−0.005	25.27
employment	(0.009)	(0.029)	(0.021)	(0.027)	(0.018)	
Overall employment	−0.005	0.042	0.007	0.047*	0.012	25.27
	(0.007)	(0.028)	(0.016)	(0.025)	(0.014)	
Log hourly wage	0.005	0.067***	0.068***	0.062***	0.063***	25.27
	(0.010)	(0.020)	(0.021)	(0.016)	(0.016)	
Panel B. Change from 2000 to 2016						
Manufacturing	−0.196***	−0.240**	−0.154***	−0.044	0.042	25.27
employment	(0.057)	(0.104)	(0.055)	(0.103)	(0.055)	
Nonmanufacturing	0.014	0.033	0.015	0.018	0.001	25.27
employment	(0.011)	(0.024)	(0.025)	(0.022)	(0.021)	
Overall employment	−0.006	0.017	0.006	0.023	0.012	25.27
	(0.007)	(0.019)	(0.017)	(0.018)	(0.015)	
Log hourly wage	0.012	0.084***	0.065***	0.072***	0.053***	25.27
	(0.012)	(0.027)	(0.020)	(0.021)	(0.014)	

Notes: The table applies the Kahn–Oldenski–Park trade specification to the 130 Great Migration commuting zones. Chinese import exposure and its race interactions are instrumented with import growth in other high-income countries and the corresponding interactions. The first three columns report group-specific effects; the next two compare Black and Hispanic workers with White workers. Employment outcomes are changes in employment relative to the working-age population; wage outcomes are changes in log hourly wages. Regressions include demographic, occupational-task, region, and race-interacted controls and use race-specific 2000 working-age-population weights. Standard errors are clustered by state. Each specification contains 389 or 390 commuting-zone-by-race observations. The first-stage statistic is the smallest of the three Sanderson–Windmeijer conditional F statistics, computed at the industry level as in Table 2.

Table A7: Import competition and mortality among men with high school or less, 2000–2013

	All men		White men		Black men	
	No interaction (1)	Interacted (2)	No interaction (3)	Interacted (4)	No interaction (5)	Interacted (6)
Panel A. Accidental drug poisoning						
China import shock (one SD)	24.4	35.1	33.0	48.0	20.3	23.1
	(30.8)	(28.0)	(35.0)	(33.1)	(35.6)	(25.7)
China shock × Predicted	–	163.8***	–	198.1***	–	118.2**
migration (one IQR)		(58.0)		(71.1)		(51.6)
Panel B. Suicide, alcoholic liver disease, and drug poisoning						
China import shock (one SD)	45.7	48.5	48.5	57.9	32.8	33.6
	(30.3)	(31.6)	(32.5)	(37.3)	(50.1)	(37.6)
China shock × Predicted	–	136.5**	–	163.4**	–	143.3*
migration (one IQR)		(60.3)		(83.1)		(82.7)
First-stage F: China shock	88.05	47.12	86.65	46.58	92.81	52.20
First-stage F: interaction	–	52.42	–	53.06	–	53.86
SW conditional F: shock	–	39.53	–	38.53	–	42.83
SW conditional F: interaction	–	39.08	–	38.07	–	42.81
Commuting zones	130	130	130	130	129	129
Industry clusters	136	136	136	136	136	136

Notes: Outcomes are cumulative deaths per 100,000 men ages 20–39 with high school or less over 2000–2013. The narrow outcome is accidental drug poisoning. The broader outcome adds suicide, alcoholic liver disease, and drug poisoning of undetermined intent. The columns and scaling follow Table 2. Deaths with unknown education are allocated according to known-education death shares within commuting zone, year, race, and cause. Fixed 1990 male education shares for ages 18–44 divide annual population estimates for ages 20–39. Each model controls for its own race-by-education all-cause mortality over 1990–1999 and uses 2000 population weights. One commuting zone lacks a usable Black education denominator, leaving 129 Black destinations and a median predicted migration rank of 51. First-stage statistics are defined as in Table 2. Exposure-robust standard errors in parentheses are clustered by three-digit industry. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Import competition and mortality among men with education beyond high school, 2000–2013

	All men		White men		Black men	
	No interaction (1)	Interacted (2)	No interaction (3)	Interacted (4)	No interaction (5)	Interacted (6)
Panel A. Accidental drug poisoning						
China import shock (one SD)	7.4	8.5	6.7	8.4	−3.1	−5.9
	(8.0)	(7.2)	(8.6)	(8.2)	(8.4)	(8.6)
China shock × Predicted	–	43.0**	–	43.6**	–	15.5
migration (one IQR)		(18.0)		(19.4)		(21.1)
Panel B. Suicide, alcoholic liver disease, and drug poisoning						
China import shock (one SD)	8.6	5.5	8.1	3.8	15.8	9.6
	(8.2)	(8.2)	(8.7)	(9.1)	(13.2)	(13.2)
China shock × Predicted	–	14.3	–	4.9	–	19.5
migration (one IQR)		(16.8)		(17.9)		(27.4)
First-stage F: China shock	86.78	45.93	84.69	45.36	97.54	55.32
First-stage F: interaction	–	50.73	–	51.50	–	61.52
SW conditional F: shock	–	38.13	–	36.95	–	40.97
SW conditional F: interaction	–	37.63	–	36.37	–	40.83
Commuting zones	130	130	130	130	129	129
Industry clusters	136	136	136	136	136	136

Notes: Outcomes are cumulative deaths per 100,000 men ages 20–39 with education beyond high school over 2000–2013. The narrow outcome is accidental drug poisoning. The broader outcome adds suicide, alcoholic liver disease, and drug poisoning of undetermined intent. The columns and scaling follow Table 2. Deaths with unknown education are allocated according to known-education death shares within commuting zone, year, race, and cause. Fixed 1990 male education shares for ages 18–44 divide annual population estimates for ages 20–39. Each model controls for its own race-by-education all-cause mortality over 1990–1999 and uses 2000 population weights. One commuting zone lacks a usable Black education denominator, leaving 129 Black destinations and a median predicted migration rank of 51. First-stage statistics are defined as in Table 2. Exposure-robust standard errors in parentheses are clustered by three-digit industry. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Unknown education and the accidental drug poisoning results, 2000–2013

	Exclude unknown		Proportional allocation	
Group	Effect (SE)	p	Effect (SE)	p
A. High school or less: effect at the 75th percentile				
All men	100.6** (41.9)	0.016	105.5** (43.7)	0.016
White men	122.6** (47.6)	0.010	133.2*** (51.2)	0.009
Black men	69.6** (35.3)	0.049	72.9* (38.5)	0.058
B. Beyond high school: effect at the 75th percentile				
All men	25.8** (11.2)	0.022	27.1** (13.0)	0.037
White men	27.0** (12.3)	0.029	27.1* (14.2)	0.056
Black men	-0.3 (14.5)	0.983	0.6 (15.5)	0.968
C. Difference between education groups at the 75th percentile				
All men	74.8** (32.1)	0.020	78.5** (32.8)	0.017
White men	95.6*** (37.0)	0.010	106.0*** (39.1)	0.007
Black men	69.9** (27.5)	0.011	72.2** (28.6)	0.011
D. High school or less: 75th-minus-25th-percentile difference				
All men	161.7*** (56.3)	0.004	163.8*** (58.0)	0.005
White men	189.8*** (68.3)	0.005	198.1*** (71.1)	0.005
Black men	119.8** (47.2)	0.011	118.2** (51.6)	0.022

Notes: Entries compare mortality effects when deaths with unknown education are excluded with effects when these deaths are allocated in proportion to known-education deaths within commuting zone, year, race, and cause. Cells without known-education deaths use the corresponding pooled 1990–2013 shares, then 1990 population shares if needed. Allocation applies to outcome deaths and education-specific lagged all-cause mortality. Both specifications use the same fixed 1990 education shares to divide annual population denominators. Effects are for a one-standard-deviation China shock on cumulative deaths per 100,000 men ages 20–39 over 2000–2013. All-men and White estimates use 130 commuting zones; Black estimates use 129. Education differences account for covariance between equations; fitted effects account for covariance between the main and interaction terms. Exposure-robust standard errors are clustered by three-digit industry. The reported p-values are two-sided normal-reference tests. Stars use unrounded p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Unknown education and the broader mortality results, 2000–2013

	Exclude unknown		Proportional allocation	
Group	Effect (SE)	p	Effect (SE)	p
A. High school or less: effect at the 75th percentile				
All men	87.8* (45.9)	0.056	107.1** (45.6)	0.019
White men	109.6** (53.4)	0.040	128.1** (56.4)	0.023
Black men	79.6** (39.4)	0.043	93.9** (38.2)	0.014
B. Beyond high school: effect at the 75th percentile				
All men	8.3 (10.5)	0.430	11.6 (13.1)	0.374
White men	5.3 (11.5)	0.648	5.9 (13.4)	0.660
Black men	6.2 (22.4)	0.783	17.9 (23.1)	0.439
C. Difference between education groups at the 75th percentile				
All men	79.5** (38.5)	0.039	95.5** (38.4)	0.013
White men	104.4** (44.8)	0.020	122.2*** (46.5)	0.009
Black men	73.4** (33.9)	0.031	76.0** (31.1)	0.014
D. High school or less: 75th-minus-25th-percentile difference				
All men	138.7** (68.8)	0.044	136.5** (60.3)	0.024
White men	162.8* (86.7)	0.060	163.4** (83.1)	0.049
Black men	143.7* (83.0)	0.083	143.3* (82.7)	0.083

Notes: Entries compare mortality effects when deaths with unknown education are excluded with effects when these deaths are allocated in proportion to known-education deaths within commuting zone, year, race, and cause. Cells without known-education deaths use the corresponding pooled 1990–2013 shares, then 1990 population shares if needed. Allocation applies to outcome deaths and education-specific lagged all-cause mortality. Both specifications use the same fixed 1990 education shares to divide annual population denominators. Effects are for a one-standard-deviation China shock on cumulative deaths per 100,000 men ages 20–39 over 2000–2013. All-men and White estimates use 130 commuting zones; Black estimates use 129. Education differences account for covariance between equations; fitted effects account for covariance between the main and interaction terms. Exposure-robust standard errors are clustered by three-digit industry. The reported p-values are two-sided normal-reference tests. Stars use unrounded p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Population responses to Chinese import competition, 2000–2012

Population outcome	Effect at 75th percentile	75th-minus-25th percentile
All people, ages 18–39	−0.575	2.001
	(1.726)	(2.144)
All men, ages 18–39	−0.606	2.334
	(1.731)	(2.180)
White men, ages 18–39	0.113	3.064
	(2.023)	(2.541)
Nonwhite men, ages 18–39	−2.282	0.808
	(1.950)	(2.345)
All men, ages 18–24	0.298	3.994**
	(1.240)	(1.761)
All men, ages 25–39	−1.236	1.422
	(2.947)	(3.701)
First-stage F: China shock	21.32	
First-stage F: interaction	22.18	
SW conditional F: shock	31.53	
SW conditional F: interaction	31.20	

Notes: Outcomes are 100 times the change in log population between 2000 and 2012. The first column gives the effect of a one-standard-deviation China shock at the 75th percentile of predicted migration; the second gives the change in that effect between the 25th and 75th percentiles. Regressions use the main trade exposure, instruments, controls, and 2000 population weights, and additionally control for the 1970–1990 population trend. The population series follow Autor et al. (2021); ages 18–39 are the closest available group to the mortality sample, and the nonwhite category is broader than Black. Net population changes cannot identify gross migration or health selection among movers. First-stage statistics are defined as in Table 2. Exposure-robust standard errors in parentheses are clustered by three-digit industry. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Mortality interactions under randomized migration assignments, 2000–2013

Model	Observed statistic	Extreme draws	Successful draws	Randomization p
Panel A. Accidental drug poisoning				
Structural IV	2.622	6	1,000	0.007
Reduced form	2.239	38	1,000	0.039
Panel B. Suicide, alcoholic liver disease, and drug poisoning				
Structural IV	1.902	0	1,000	0.001
Reduced form	1.666	8	1,000	0.009

Notes: The observed statistic is the all-men China-shock interaction coefficient divided by its exposure-robust standard error. Each of 1,000 draws replaces predicted Great Migration with a randomized migration assignment from Derenoncourt (2022), including every interaction in which it appears. The mortality outcome, China-shock exposure and instrument, other controls, weights, and 130-zone sample are held fixed. Extreme draws are those with a statistic at least as large in absolute value as the observed statistic. Two-sided randomization p-values add one to the number of extreme draws and to the number of draws. The narrow outcome is accidental drug poisoning. The broader outcome adds suicide, alcoholic liver disease, and drug poisoning of undetermined intent.

Table A13: Mortality effects with education, health, and occupation controls, 2000–2013

	Accidental drug poisoning		Broader mortality	
Added controls	White men	Black men	White men	Black men
Panel A. Effect at the 75th percentile				
Baseline	73.9**	46.4	68.1**	75.4**
	(30.1)	(33.6)	(30.8)	(38.2)
Education industry, 2000	73.8**	46.4	68.0**	75.4**
	(30.1)	(31.9)	(31.3)	(34.2)
Health/social assistance, 2000	73.9**	42.7	70.1**	72.5**
	(30.4)	(30.3)	(32.3)	(35.3)
Education + health, 2000	73.8**	42.7	70.0**	72.4**
	(30.3)	(28.9)	(32.2)	(31.6)
Occupations, 1990 proxy	54.6**	47.5*	56.7*	70.5**
	(26.6)	(28.5)	(29.6)	(28.2)
Occupations, 2000	43.4*	37.0	41.3	48.4*
	(25.3)	(28.5)	(28.1)	(27.2)
Industries + occupations, 2000	52.6	38.0	64.7*	66.0**
	(35.1)	(24.1)	(36.6)	(28.3)
Panel B. 75th-minus-25th-percentile difference				
Baseline	105.4***	81.7**	69.3*	97.7*
	(38.8)	(41.0)	(37.9)	(54.4)
Education industry, 2000	109.6***	83.9**	75.8**	103.3**
	(37.1)	(39.9)	(34.9)	(50.5)
Health/social assistance, 2000	105.6**	61.3	80.1*	81.6
	(41.7)	(43.4)	(41.6)	(57.6)
Education + health, 2000	109.7***	63.2	86.6**	86.9
	(39.7)	(42.2)	(38.7)	(53.0)
Occupations, 1990 proxy	68.9*	63.4*	57.5	75.3*
	(39.3)	(38.4)	(38.7)	(45.2)
Occupations, 2000	85.9***	65.6*	52.3*	68.6*
	(31.2)	(34.9)	(31.4)	(37.1)
Industries + occupations, 2000	101.9***	50.0*	116.6**	85.4*
	(39.0)	(30.0)	(45.6)	(44.3)

Notes: The indicated industry or occupation shares are added to the main specification, retaining its 130 commuting zones, instruments, baseline controls, and population weights. Panel A evaluates the China-shock effect at the 75th percentile of predicted migration; Panel B reports the difference between the 75th and 25th percentiles. Effects are for a one-standard-deviation China shock, in cumulative deaths per 100,000 men ages 20–39 over 2000–2013. The narrow outcome is accidental drug poisoning. The broader outcome adds suicide, alcoholic liver disease, and drug poisoning of undetermined intent. The 2000 shares describe employed civilian residents ages 16 and older, including public employment. Education and health/social assistance are industries; occupation shares describe types of work. The 1990 occupation measures use county-population-weighted Census proxies. These controls enter additively. Changes in coefficients do not identify a causal contribution of public services or occupations. Exposure-robust standard errors in parentheses are clustered by three-digit industry. Stars use unrounded p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A14: Trade liberalization and mortality: age coverage and pretrend tests

Group	Ages	Baseline estimate (SE)	Pretrend p-value	With slopes estimate (SE)	Pretrend p-value
Panel A. Accidental drug poisoning					
All men	20–39	6.57** (2.47)	0.132	7.55*** (2.16)	0.049
White men	20–39	7.35*** (2.50)	0.153	8.19*** (2.20)	0.056
All men	20–64	5.29** (2.20)	0.002	7.57*** (1.85)	0.002
White men	20–64	5.43** (2.38)	< 0.001	7.00*** (2.44)	< 0.001
All men	All ages	3.40** (1.30)	< 0.001	4.84*** (1.15)	< 0.001
White men	All ages	3.36** (1.43)	< 0.001	4.29*** (1.52)	< 0.001
Panel B. Broader mortality					
All men	20–39	7.63* (4.04)	< 0.001	5.57 (4.97)	0.007
White men	20–39	8.21** (3.64)	< 0.001	7.19* (3.72)	0.016
All men	20–64	5.94* (3.08)	< 0.001	6.00 (3.59)	< 0.001
White men	20–64	5.54* (2.73)	< 0.001	5.45* (2.81)	< 0.001
All men	All ages	3.68** (1.72)	0.004	3.94* (2.00)	0.001
White men	All ages	3.23** (1.53)	0.001	3.26* (1.61)	< 0.001
Panel C. All-cause mortality					
All men	20–39	18.79** (7.21)	0.041	17.12*** (6.20)	0.068
White men	20–39	11.39 (8.07)	0.013	10.36 (7.12)	0.008
All men	20–64	19.39 (14.45)	< 0.001	12.13 (9.68)	< 0.001
White men	20–64	9.41 (14.09)	< 0.001	4.81 (10.89)	< 0.001
All men	All ages	6.19 (12.84)	< 0.001	-0.20 (11.63)	< 0.001
White men	All ages	-2.50 (15.28)	< 0.001	-6.59 (15.60)	< 0.001

Notes: Entries average the 2008–2013 interaction coefficients relative to 1999. They measure the difference in the PNTR mortality response across one interquartile range of predicted Great Migration, per interquartile range of tariff-gap exposure. Outcomes are annual age-standardized deaths per 100,000, using common national population shares in 2000 renormalized within each age range. Broader mortality combines accidental and undetermined drug poisoning, suicide, and alcoholic liver disease; all-cause mortality includes every cause. The baseline adapts Pierce–Schott controls: Chinese tariff and subsidy changes, initial veteran and foreign-born shares and manufacturing employment interacted with year, plus annual applied tariffs and textile-quota exposure. Their original income and education controls are omitted. Both models include commuting-zone and year effects and both lower-order exposure-by-year terms. The second adds group- and outcome-specific 1990–1999 mortality slopes interacted with year. Each regression contains 3,120 observations from 130 zones and uses fixed 1990 age- and group-specific population weights. State-clustered standard errors appear in parentheses (32 states); slope-model inference conditions on estimated slopes. Pretrend p-values jointly test the nine 1990–1998 interaction coefficients relative to 1999. Stars are pointwise; age ranges overlap. These annual reduced forms differ from the main cumulative instrumental-variable effects. Stars use unrounded p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A15: The Great Migration, segregation, and concentrated poverty in 1990

Outcome	IV effect (SE)	AR 95% set	AR p	Mean	CZs	First-stage F
Panel A. Racial segregation						
White–Black dissimilarity (LTDB)	4.96*** (1.23)	[2.988, 10.037]	<0.001	59.7	130	9.6
Black isolation (LTDB)	9.55*** (1.94)	[6.782, 18.512]	<0.001	24.6	130	9.6
Black exposure to White residents (LTDB)	−9.15*** (1.70)	[−16.284, −6.455]	<0.001	66.2	130	9.6
Racial Theil index, 2000 (Derenoncourt)	3.80*** (0.91)	[2.472, 7.930]	<0.001	19.3	130	9.6
Panel B. Segregation by poverty and education						
Poor–nonpoor dissimilarity, all residents	2.43*** (0.65)	[1.165, 4.634]	0.004	31.5	130	9.6
Income segregation, 2000 (Derenoncourt)	0.53** (0.24)	[−0.196, 1.041]	0.100	7.1	130	9.6
Poor–nonpoor dissimilarity, White	1.69*** (0.58)	[0.582, 3.701]	0.009	28.1	130	9.6
Poor–nonpoor dissimilarity, Black	−4.16*** (1.33)	[−8.739, −1.470]	0.011	40.1	124	8.9
College–noncollege dissimilarity, White (NHGIS)	1.52*** (0.49)	[0.385, 2.893]	0.021	28.4	130	9.6
College–noncollege dissimilarity, Black (NHGIS)	−5.31*** (1.57)	[−10.850, −2.567]	0.001	50.5	128	10.8
Panel C. Concentrated poverty (percent of poor in ≥40% poverty tracts)						
All poor residents	3.56*** (1.07)	[1.942, 8.216]	<0.001	8.9	130	9.6
White poor residents	1.82** (0.87)	[0.412, 5.377]	0.012	5.1	130	9.6
Black poor residents	8.05*** (2.29)	[4.559, 18.501]	<0.001	19.0	124	8.9
Memo: CZ poverty rate, all	0.54 (0.43)	[−0.147, 2.278]	0.129	11.8	130	9.6

Notes: This table reports the estimated effect of the Great Migration on neighborhood segregation and concentrated poverty in 1990. The unit of observation is a CZ. Each row reports a separate two-stage least squares regression. The dependent variable is the listed measure, on a 0–100 scale; the two measures from Derenoncourt (2022) refer to 2000. The independent variable is the percentile of Black population increase during the Great Migration (1940–1970), and coefficients give the effect of a ten-percentile increase. The instrument is the percentile of predicted Black population increase, defined as the interaction between pre-1940 Black southern migration patterns and post-1940 outflows of migrants as predicted by southern economic factors alone. All regressions include the baseline controls of Derenoncourt (2022): educational upward mobility in 1940, the 1940 manufacturing share of the labor force, the share of the 1940 urban population consisting of Black migrants from the South during 1935–1940, and Census region fixed effects. CZs are weighted equally. Heteroskedasticity-robust standard errors are in parentheses; Anderson–Rubin confidence sets and p-values are robust to weak instruments. Tract measures use harmonized 2010 boundaries; poverty measures use residents for whom poverty status is recorded. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Sources: Longitudinal Tract Database (LTDB), NHGIS 1990 tract tables, Derenoncourt (2022).

Table A16: Channel tests: migration gradients and shock-by-condition effects, 2000–2013

Panel A. Accidental drug poisoning

	White men		Black men		
Destination condition	Gradient without → with (W→W/)	Shock × condition (S)	Gradient without → with (W→W/)	Shock × condition (S)	Minimum first-stage F
H1. Divided polities and public goods (Trounstine)					
Racial segregation (White–Black Theil)	110→77	18.3 (30.6)	85→-5	50.0* (25.8)	62.8
White opposition to redistribution (CCES, 2014–20)	111→91	25.6* (13.2)	80→63	21.9*** (7.3)	32.8
City-manager government, 1996	98→108	14.0 (14.3)	76→72	-6.1 (8.7)	84.8
Local governments per capita, 1992 (log)	135→137	31.8 (25.9)	89→90	15.6 (16.2)	44.5
Local spending per resident, 1992 (log)	138→204	64.6 (65.4)	86→83	-2.5 (34.0)	20.8
H2. Segregation concentrates poverty (Massey)					
Poor residents in 40%+ poverty tracts, all	93→78	11.8 (19.6)	72→26	36.1*** (14.0)	55.3
Poor residents in 40%+ poverty tracts, Black	102→96	4.7 (23.7)	83→31	43.2*** (16.0)	86.1
Poor residents in 40%+ poverty tracts, White	95→94	1.8 (13.4)	64→59	8.0 (15.2)	42.7
Class segregation, all residents (Theil)	103→71	26.2 (35.6)	84→11	60.6*** (22.7)	109.6
H3. Isolation and middle-class exit (Wilson)					
Isolation of poor Black residents	118→119	-0.3 (14.8)	85→18	41.0*** (14.5)	269.1
Isolation of poor White residents	92→101	-16.7 (11.1)	66→60	11.6 (12.2)	79.5
Class segregation, Black (Theil)	104→124	24.3* (12.4)	82→92	12.6 (10.2)	65.8
Education segregation, Black (Theil)	109→111	4.2 (17.6)	87→92	12.3 (11.7)	100.9
Class segregation, White (Theil)	106→82	30.6 (46.5)	83→56	34.3 (26.2)	31.4
Education segregation, White (Theil)	115→103	24.6 (24.2)	86→71	29.9* (16.9)	60.3
Other destination conditions					
Education segregation, all residents (Theil)	110→99	26.7 (20.8)	82→70	27.6* (14.9)	71.4
Prison rate, 1983–2000	110→100	29.5 (52.0)	103→93	28.7 (38.4)	29.9
Routine-occupation share	93→103	-8.5 (23.1)	72→31	34.9** (16.0)	53.8
Routine task intensity, manufacturing	110→106	6.3 (4.9)	70→58	16.8*** (5.8)	52.9

Notes: "W→W/" is the gradient without and with the shock-by-condition term (same sample and controls). "Shock×S" is the change in the effect of a one-SD China shock, in cumulative 2000–2013 deaths per 100,000 men ages 20–39, per one-SD higher condition; exposure-robust standard errors in parentheses. F is the smallest BHJ-form Sanderson–Windmeijer first-stage statistic among the three endogenous terms. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A16: Channel tests: migration gradients and shock-by-condition effects, 2000–2013 (continued)

Panel B. Suicide, alcoholic liver disease, and drug poisoning

	White men		Black men		
Destination condition	Gradient without → with (W→W/)	Shock × condition (S)	Gradient without → with (W→W/)	Shock × condition (S)	Minimum first-stage F
H1. Divided polities and public goods (Trounstine)					
Racial segregation (White–Black Theil)	75→110	-19.7 (28.8)	96→88	4.6 (37.9)	62.8
White opposition to redistribution (CCES, 2014–20)	75→66	10.9 (12.9)	100→96	5.4 (11.3)	32.8
City-manager government, 1996	69→76	10.1 (13.8)	96→92	-6.6 (10.7)	84.8
Local governments per capita, 1992 (log)	86→86	-0.8 (29.4)	114→113	-13.0 (40.5)	44.5
Local spending per resident, 1992 (log)	76→101	24.2 (29.7)	104→83	-20.0 (41.2)	20.8
H2. Segregation concentrates poverty (Massey)					
Poor residents in 40%+ poverty tracts, all	58→93	-27.4 (17.6)	84→85	-0.7 (21.7)	55.3
Poor residents in 40%+ poverty tracts, Black	66→115	-40.6 (25.0)	98→93	3.9 (26.7)	86.1
Poor residents in 40%+ poverty tracts, White	56→66	-17.6 (12.1)	73→72	2.3 (22.2)	42.7
Class segregation, all residents (Theil)	83→85	-1.4 (31.7)	114→78	30.3 (30.9)	109.6
H3. Isolation and middle-class exit (Wilson)					
Isolation of poor Black residents	82→138	-33.9** (15.1)	100→82	10.7 (23.3)	269.1
Isolation of poor White residents	58→77	-37.8** (15.0)	84→81	4.8 (22.4)	79.5
Class segregation, Black (Theil)	67→84	20.9 (18.5)	99→109	11.5 (25.5)	65.8
Education segregation, Black (Theil)	77→76	-4.5 (19.6)	112→120	19.3 (29.1)	100.9
Class segregation, White (Theil)	71→71	0.1 (42.0)	97→82	19.1 (33.0)	31.4
Education segregation, White (Theil)	74→68	13.9 (25.6)	92→85	16.1 (27.3)	60.3
Other destination conditions					
Education segregation, all residents (Theil)	74→66	19.5 (21.7)	92→84	17.4 (22.6)	71.4
Prison rate, 1983–2000	98→96	3.0 (39.9)	129→124	13.5 (39.4)	29.9
Routine-occupation share	61→61	-0.4 (22.7)	87→31	47.7** (20.4)	53.8
Routine task intensity, manufacturing	70→68	2.7 (5.6)	78→65	18.0** (7.0)	52.9

Notes: "W→W/" is the gradient without and with the shock-by-condition term (same sample and controls). "Shock×S" is the change in the effect of a one-SD China shock, in cumulative 2000–2013 deaths per 100,000 men ages 20–39, per one-SD higher condition; exposure-robust standard errors in parentheses. F is the smallest BHJ-form Sanderson–Windmeijer first-stage statistic among the three endogenous terms. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A17: Local public spending, 2000–2007: the Great Migration gradient and White opposition to redistribution

	Great Migration gradient		Shock × White opposition		Gradient, 2000–07	
Outcome (units)	2000–07	1990–2000	Alone	With GM	Without	With
Total expenditure (percent)	−0.2	−0.3	1.3	1.3	−1.8	−2.9
	(4.8)	(1.6)	(1.0)	(1.1)		
Education (percent)	4.1	−4.1*	2.6**	2.6**	2.5	0.5
	(4.0)	(2.4)	(1.3)	(1.3)		
Public safety (police, fire, corrections) (percent)	0.4	0.2	1.3	1.3	−1.7	−2.7
	(6.4)	(3.8)	(1.4)	(1.4)		
Health and hospitals ($ per resident)	−78.6**	19.4	−30.7***	−30.0***	−67.8	−44.5
	(36.2)	(13.3)	(7.0)	(6.5)		
Public welfare and housing ($ per resident)	−21.8	25.3*	9.5	10.0	−25.4	−33.3
	(23.7)	(15.2)	(8.4)	(8.1)		
Own-source revenue (percent)	−4.2	8.5	2.1*	2.2*	−6.8	−8.5
	(6.7)	(6.0)	(1.1)	(1.3)		
State intergovernmental transfers (percent)	4.1	−14.2*	1.3	1.3	2.9	1.9
	(7.7)	(8.0)	(1.5)	(1.6)		

Notes: Outcomes are changes in local government spending and revenue per resident in each commuting zone, in inflation-adjusted dollars (log changes × 100, in percent, where the category has no zeros), from the Census of Governments as assembled by Feler and Senses (2017); 2000–2007 changes are scaled to a decade, as in their paper. The 1990–2000 column repeats the gradient for the decade before the shock. Each regression uses the specification of Table 2 (130 commuting zones, population weights, exposure-robust standard errors in parentheses) with the spending change as the outcome. "Great Migration gradient" is the difference in the effect of a one-standard-deviation China shock between the 75th and 25th percentiles of predicted Great Migration rank. "Shock × White opposition" is the change in that effect per standard deviation of White residents' opposition to redistributive spending (CCES, 2014–2020), alone and jointly with the Great Migration interaction. The last two columns report the gradient on the same sample and controls without and with the opposition interaction. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

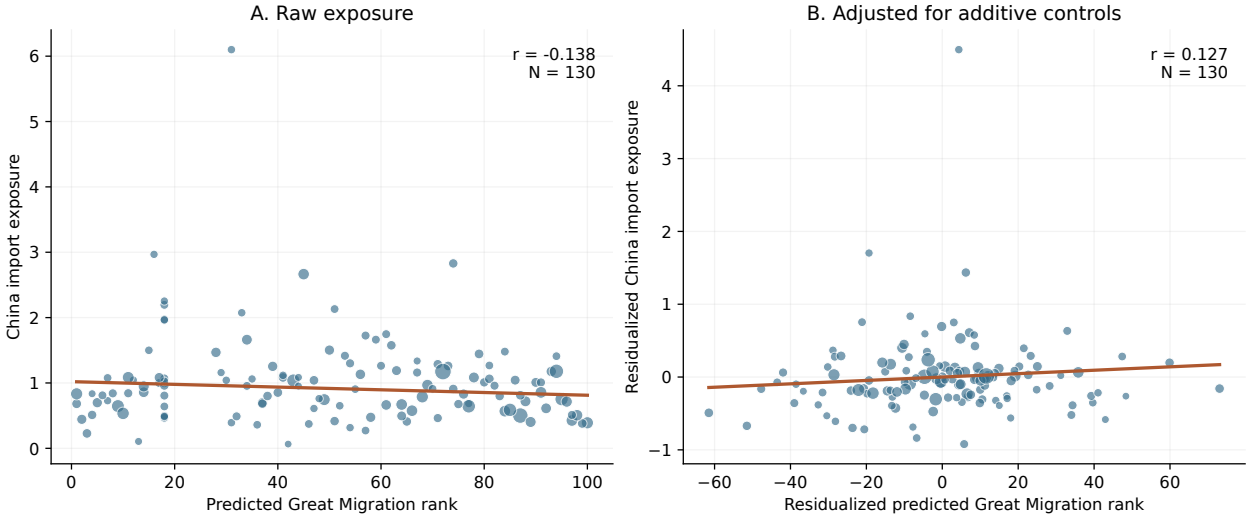
Table A18: The China shock in urban-core counties and the rest of the commuting zone, 2000–2013

	Accidental drug poisoning			Broader outcome		
	All	White	Black	All	White	Black
2000–2012 shock; 2000–2013 mortality						
Whole CZ, at p75	65.8**	73.9**	46.4	60.7**	68.1**	75.4**
	(27.9)	(30.1)	(33.6)	(27.2)	(30.8)	(38.2)
Whole CZ, gradient	98.1***	105.4***	81.7**	68.3*	69.3*	97.7*
	(37.4)	(38.8)	(41.0)	(35.9)	(37.9)	(54.4)
Core county, at p75	99.2***	123.7***	58.5	107.6***	128.1***	121.8**
	(32.1)	(39.1)	(46.9)	(32.8)	(42.0)	(57.3)
Core county, gradient	138.3***	154.5***	134.1**	91.1**	81.6*	181.7**
	(39.5)	(42.5)	(60.6)	(38.1)	(46.6)	(78.1)
Rest of CZ, at p75	38.0	48.1	-9.1	43.5	60.0	-49.3
	(42.1)	(44.5)	(29.3)	(41.1)	(44.3)	(50.7)
Rest of CZ, gradient	75.1	95.0	3.5	47.0	83.5	-92.0
	(65.0)	(67.7)	(44.1)	(69.8)	(70.7)	(78.6)
Rest − core, at p75	-62.1	-77.8	-69.3**	-66.0*	-71.6	-172.8***
	(39.7)	(48.9)	(33.1)	(38.1)	(47.6)	(64.1)
Rest − core, gradient	-85.7	-86.5	-154.6***	-56.6	-9.1	-300.2***
	(62.6)	(69.7)	(57.3)	(72.8)	(76.6)	(115.6)
1990–2000 shock; 1990–1999 mortality						
Whole CZ, at p75	-9.7	-15.2	-6.3	14.4	-7.3	98.9**
	(17.6)	(15.1)	(33.8)	(15.7)	(13.2)	(41.5)
Whole CZ, gradient	-8.4	-14.4	7.0	10.5	-8.5	70.6*
	(22.1)	(18.8)	(43.1)	(17.1)	(15.4)	(36.2)
Core county, at p75	-27.1	-40.9	-9.7	45.9	1.9	129.8**
	(34.5)	(39.1)	(38.3)	(40.7)	(38.4)	(64.5)
Core county, gradient	-34.8	-52.4	6.7	17.1	-26.3	66.7
	(40.5)	(44.5)	(51.4)	(42.7)	(39.9)	(52.7)
Rest of CZ, at p75	-14.4*	-14.3	-2.2	-15.8	-20.1	1.4
	(8.5)	(8.7)	(33.3)	(17.1)	(17.8)	(44.4)
Rest of CZ, gradient	0.4	0.1	-19.4	8.8	5.8	14.6
	(12.2)	(12.7)	(54.6)	(24.0)	(24.7)	(70.7)
Rest − core, at p75	10.1	23.6	2.6	-69.2	-28.2	-134.0**
	(30.8)	(36.4)	(34.5)	(43.5)	(41.8)	(66.3)
Rest − core, gradient	25.0	46.1	-39.4	-19.0	28.4	-52.9
	(38.3)	(44.4)	(53.5)	(55.0)	(51.1)	(97.8)

Notes: This table reports the effect of the China shock on mortality separately for each CZ's urban-core county and the rest of the CZ. The unit of observation is a CZ. The core is the county with the most principal-city census tracts (2010 definitions), plus any county lying entirely within a principal city, so that the five New York City boroughs form one core; the rest of the CZ comprises all other counties. The dependent variable is the cumulative death rate per 100,000 men ages 20–39 in the indicated area, built from county death counts and SEER populations. Rows report the effect of a one-standard-deviation shock at the 75th percentile of predicted Great Migration rank and the difference between the 75th and 25th percentiles; "Rest − core" estimates the difference directly. The upper panel uses the 2000–2012 China shock and 2000–2013 mortality with the specification of Table 2; core rows use the 116 CZs with a principal-city county and rest rows the 113 that also have another county. The lower panel uses the 1990–2000 China shock, 1990–1999 mortality, 1990 controls and population weights, and cumulative 1980s all-cause mortality as in Autor et al. (2019). Deaths are recorded by county of residence, so core counties include inner suburbs of large central counties. Exposure-robust standard errors, clustered by three-digit industry following Borusyak et al. (2022), are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Sources: NCHS restricted-use mortality files, SEER county populations, LTDB 2010 principal-city tract flags, Autor et al. (2021).

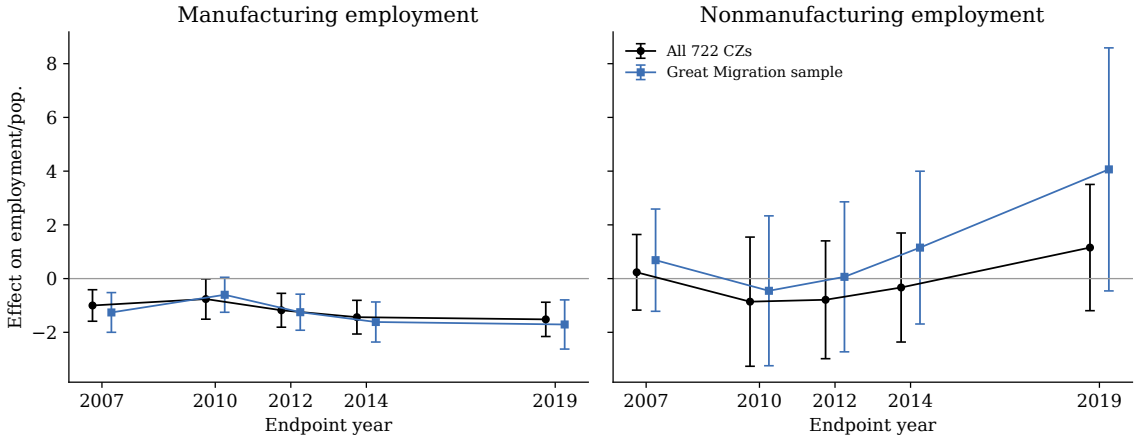
Appendix Figures

Figure A1: Predicted Great Migration and Chinese import exposure, 2000–2012



Notes: Each point represents one of the 130 commuting zones in the mortality sample. Panel A plots observed Chinese import exposure against predicted Great Migration rank. Panel B plots residuals from separate regressions of the two measures on the additive controls listed in Appendix Table A5. Residualization, fitted lines, and correlations use 2000 population weights. Marker areas increase with the square root of the population weight. Import exposure is the ten-year equivalent of the 2000–2012 change in Chinese import penetration, measured in percentage points.

Figure A2: Employment responses to Chinese import competition: national and Great Migration samples



Notes: Points report the effects of a one-unit increase in Chinese import exposure on the change in employment relative to the working-age population, from 2001 to the year indicated. Effects are in percentage points; import exposure is measured on a ten-year-equivalent scale. The left panel shows manufacturing employment and the right panel nonmanufacturing employment. Black circles represent all 722 commuting zones; blue squares represent the 130-zone Great Migration sample. Bars are 95 percent confidence intervals based on exposure-robust standard errors clustered by three-digit industry. Plot positions are offset for legibility.